

Decoding protein phosphorylation during oocyte meiotic divisions using phosphoproteomics

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eLife Assessment

This **valuable** paper describes a comprehensive quantitative phospho-proteomic analysis of *Xenopus* oocytes during meiosis. Using time-resolved proteomic analyses, the authors provide insights into changes in protein levels and phosphorylation states to an unprecedented depth, quality, and quantitative detail. The key findings are **solid** and offer a helpful resource for the scientific community.

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Abstract

Oocyte meiotic divisions represent a critical process in sexual reproduction, as a diploid non-dividing oocyte is transformed into a haploid fertilizable egg, as a prelude for the subsequent embryonic divisions and differentiation. Although cell differentiation and proliferation are governed by transcription, oocyte maturation and early embryonic divisions depend entirely on changes in protein abundance and post-translational modifications. Here, we analyze the abundance and phosphorylation of proteins during *Xenopus* oocyte meiotic maturation. We reveal significant shifts in protein stability, related to spindle assembly, DNA replication and RNA-binding. Our analysis pinpoints broad changes in phosphorylation correlating with key cytological meiotic milestones, noteworthy changes in membrane trafficking, nuclear envelope disassembly and modifications in microtubule dynamics. Additionally, specific phosphorylation events target regulators of protein translation, Cdk1 and the Mos/MAPK pathway, thereby providing insight into the dynamics of Cdk1 activity, as related to the meiotic cell cycle. This study sheds light on the orchestration of protein dynamics and phosphorylation events during oocyte meiotic divisions, providing a rich resource for understanding the molecular pathways orchestrating meiotic progression in the frog, and most likely applicable to other vertebrate species.

Introduction

In mature females, the oocyte undergoes a long period of growth, during which meiosis is arrested at prophase I. In the *Xenopus* oocyte, transcription is switched off when the oocyte reaches the end of its growth phase (Dumont, 1972 [\[1\]](#)). The full-grown oocyte is transcriptionally silent and equipped with a stock of mRNAs and proteins that will support without transcription three critical embryonic processes: first, two meiotic divisions that transform the oocyte into a fertilizable haploid egg; second, the process of fertilization; and third, 13 embryonic divisions, after which transcription is initiated.

The transformation of a fully-grown oocyte arrested in prophase into a fertilizable cell, arrested in metaphase of the second meiotic division is a process called meiotic maturation. Meiotic maturation is triggered by progesterone, secreted by the follicle cells surrounding the oocyte. The secretion of progesterone is stimulated by Luteinizing Hormone produced by the pituitary. Progesterone initiates a molecular signaling cascade that lasts 3 to 4 hours and occurs with little change in oocyte morphology. Notably, this cascade leads to the activation of the Cdk1-Cyclin B kinase, the universal inducer of cell division in eukaryotes. Cdk1 phosphorylates many substrates that promote meiotic nuclear events: nuclear envelope breakdown (NEBD), the formation of the first meiotic spindle (MI), the completion of the first meiotic division with expulsion of a polar body, and formation of the second meiotic spindle (MII) (Figure 1 [\[1\]](#)). As in all vertebrates, the frog oocyte arrests a second time in metaphase II, until fertilization stimulates the completion of the second meiotic division. The entire process of oocyte maturation occurs in the absence of transcription. The regulation of the two meiotic divisions and the 13 embryonic divisions that follow are thought to be largely regulated by changes in protein abundance and phosphorylation/dephosphorylation.

The prophase-arrested oocyte contains a store of Cdk1-Cyclin B complexes that are kept inhibited by Cdk1 phosphorylation at Y15 and T14, which are substrates of the kinase, Myt1 (Mueller *et al.*, 1995 [\[2\]](#)). In all vertebrates, the prophase arrest is maintained by high levels of cAMP and PKA (cAMP-activated protein kinase) activity (Figure 1 [\[1\]](#)). The release of this prophase block is triggered by a drop in the levels of cAMP and the consequent inhibition of PKA, which occurs within 60 min after progesterone stimulation. The identity of the critical PKA substrates whose dephosphorylation is thought to induce the pathway leading to Cdk1 activation is still unknown, with the exception of Arpp19 whose mechanism of action is not completely understood (Dupré *et al.*, 2014 [\[3\]](#); Santoni *et al.*, 2024 [\[4\]](#)). Nevertheless, a drop in PKA activity leads to the synthesis of new proteins from cytoplasmic stockpiles of mRNAs as well as increases in protein level through regulation of the ubiquitination machinery (Santoni *et al.*, 2024 [\[4\]](#)). Among the accumulated proteins are Cyclin B1, the best-known activator of Cdk1 in mitotic cells, and Mos, a kinase specific to the oocytes (Figure 1 [\[1\]](#)). The newly synthesized B1-Cyclins bind to monomeric Cdk1 to form a small pool of active complexes that evade the inhibition of Myt1 (Hocheegger *et al.*, 2001 [\[5\]](#); Gaffré *et al.*, 2011 [\[6\]](#)). Subsequently, this small amount of active Cdk1 initiates a complex network of feedback loops, involving many kinases and phosphatases, thereby creating an auto-amplification loop (Jesus, 2010 [\[7\]](#)) (Figure 1 [\[1\]](#)). Within this loop, the activation of the Cdc25 phosphatase that dephosphorylates Cdk1 and the inhibition of the PP2A phosphatase that counteracts Cdk1, lead to a rapid and full activation of Cdk1-Cyclin B (Jesus, 2010 [\[7\]](#); Lemonnier *et al.*, 2020 [\[8\]](#)). Another important player in this positive feedback loop is Mos, whose translation is induced by progesterone and which accumulates at the time of Cdk1 activation (Sagata *et al.*, 1988 [\[9\]](#), 1989a [\[10\]](#); Frank-Vaillant *et al.*, 1999 [\[11\]](#)). Mos activates Erk1/2 (also known as MAPK) indirectly, which modulates the core regulators of Cdk1 (Haccard & Jesus, 2006 [\[12\]](#)) (Figure 1 [\[1\]](#)). Cdk1 and the kinases activated under its control (Mos/MAPK, Aurora-A, Plk1, etc.) trigger a second wave of protein translation and accumulation, as well as mediating the structural changes of cell division: NEBD, chromosome condensation and formation of the Meiosis I spindle (Jesus, 2010 [\[7\]](#); Santoni

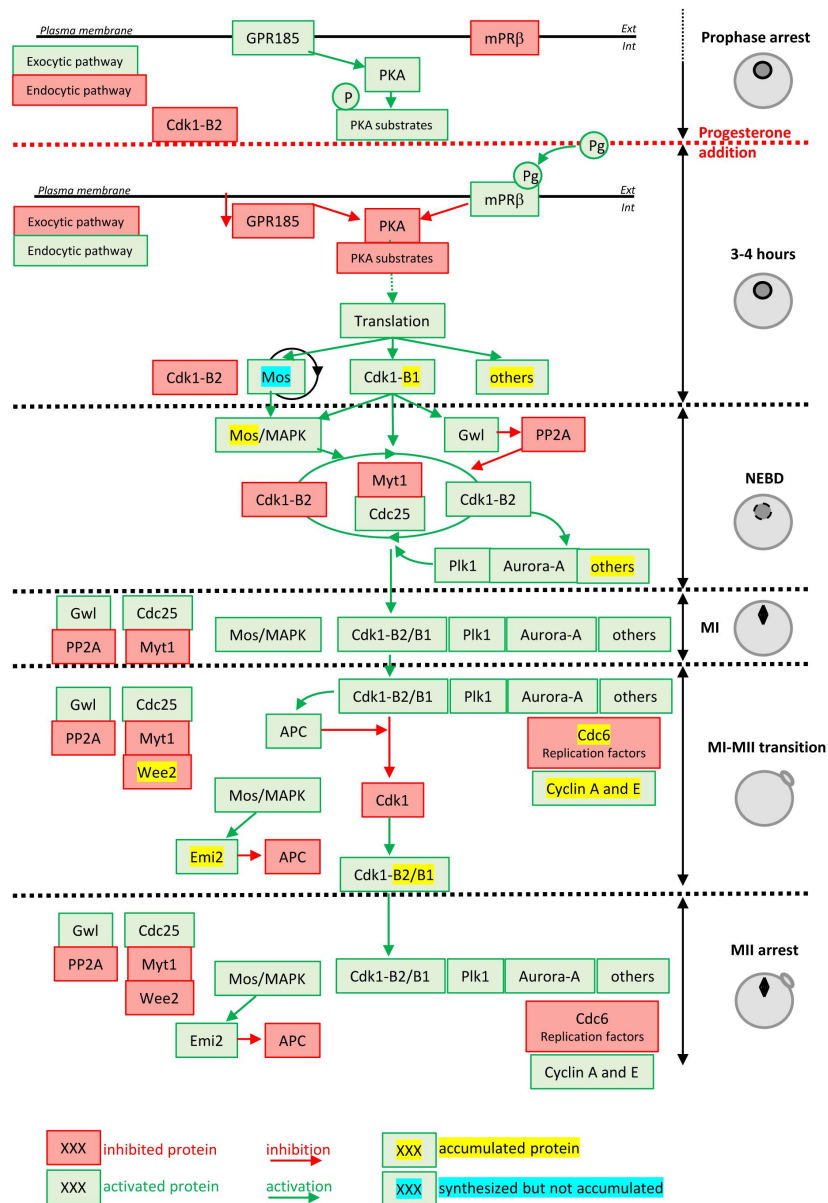


Figure 1.

Molecular events coordinating *Xenopus laevis* oocyte meiotic divisions. From top to bottom: The prophase arrest is maintained by high PKA activity due to the constitutively expressed GPR185 receptor. The exocytic pathway is active while the endocytic pathway is downregulated. Cdk1-Cyclin B2 complexes are present under an inactive state due to Cdk1 phosphorylation. Progesterone activates the mPR β receptor, GPR185 is inactivated, the exocytosis and endocytosis pathways are inverted. PKA is inactivated and promotes translation and/or accumulation of new proteins, among them Mos (that does not accumulate due to its instability) and Cyclin B1 (due to its stabilization). New Cyclin B1 proteins associate with Cdk1 and form a starter amount of Cdk1 activity. This Cdk1 starter stabilizes Mos and promotes Cdk1-Cyclin B2 activation through the regulation of Myt1, Cdc25, Gwl and PP2A. The autoamplification loop is launched, whereby more Cdk1 is activated, more it activates kinases as Mos-MAPK, Aurora-A and Plk1, all of them contributing with Cdk1 to regulate Myt1 and Cdc25, hence accelerating Cdk1 activation. Phosphorylated substrates trigger NEBD and the MI spindle assembly as well as protein translation. APC is activated under the control of Cdk1 and leads to Cyclin B degradation. The Cdk1 activity decreases and allows anaphase I and expulsion of the 1st polar body. Under the Mos/MAPK control, Emi2/Erp1 inactivates APC, halting Cyclin B degradation. The re-accumulation of Cyclin B increases Cdk1 activity that triggers the formation of the MII spindle, avoiding the assembly of a nucleus and DNA replication. APC remains inhibited by Emi2/Erp1, what ensures the oocyte arrest in MII. Color code: red: inactivated proteins; green: activated proteins; red arrow: inhibition; green arrow: activation; yellow highlight: accumulated protein; blue highlight: protein synthesized but not accumulated.

et al, 2024 [\[link\]](#)). They also activate the anaphase-promoting complex (APC), an E3 ubiquitin ligase, leading to the degradation of both Cyclin B and securin, thus enabling entry into anaphase I, chromosome separation and the expulsion of the 1st polar body (**Figure 1** [\[link\]](#)). The decrease in Cdk1 activity is rapidly followed by the inhibition of APC, which then allows the accumulation of B-Cyclins and increase of Cdk1 activity (**Figure 1** [\[link\]](#)). These events trigger the entry into metaphase II and the formation of the MII spindle. In all vertebrates, oocytes arrest in the middle of the meiotic division as a result of the stabilization of cyclin B, resulting from APC inhibition by the Emi2/Erp1 (Jesus, 2010 [\[link\]](#)) (**Figure 1** [\[link\]](#)).

The burst of phosphorylation, which occurs at NEBD when Cdk1 is activated, targets proteins belonging to three categories (Maller *et al*, 1977 [\[link\]](#)): first, regulators of translation and/or degradation of proteins that are essential for the progression of meiotic divisions; second, proteins directly involved in the intracellular reorganization of the oocyte, such as the components of meiotic spindles, the nuclear envelope or other cytoskeletal elements; third, proteins that do not have a role in meiotic divisions but regulate fertilization and early embryonic development. Notably, the sperm does not contribute any proteins or RNA except for its genome and its two centrioles. With fertilization there is extrusion of the 2nd polar body, the fusion of the male and female pronuclei. Within a few minutes after fertilization, the first S-phase starts, followed by the first 13 embryonic division cycles, all occurring in the absence of transcription. Thus, these events depend exclusively on maternal proteins and mRNA. Phosphoregulation is a central and conserved mechanism that enables remodeling of the oocyte proteome and supports cell division. It prepares the oocyte to transform into the egg and undertake development. In the absence transcription our knowledge of the dynamic patterns of phosphorylation of thousands of oocyte phosphosites represents an essential tool for study of oogenesis, embryogenesis, especially the unique meiotic and mitotic processes. Interestingly, three recent studies based on phosphoproteomics have revealed new insights into how phosphorylation dynamics regulate meiotic divisions in yeast (Celebic *et al*, 2024 [\[link\]](#); Koch *et al*, 2024 [\[link\]](#)) and mouse (Sun *et al*, 2024). We now extend this analysis to the physiological process of oogenesis specific to non-mammalian metazoans. The study of mitotic processes in somatic cells, which are short in duration, requires exquisite synchronization, which is difficult. Although previous studies of protein phosphorylation during mitosis have revealed the identity of numerous players of cell division, especially those controlling the structural reorganization of the dividing cell and its checkpoints, they have not been able to fully elucidate the epistatic relationships between kinases and phosphatases, which are nonetheless crucial to understanding the temporal sequences regulating the various events of mitosis. This weakness is due to the fact that cell proliferation is a continuous phenomenon, without physiological arrest. Its duration varies from one cell to another within the same population. The necessity of using synchronizing agents induces non-physiological arrests, which activate checkpoints, leading to the resumption of the cell cycle not comparable to that of untreated cells. Moreover, these studies often use transformed cell lines with accumulation of mutations that makes it hard to compare with cells in a physiological cellular context. The exceptional features of oocyte maturation for experimental studies of mitosis and meiosis are: 1) the synchronicity of the process in the oocyte, initiated by a physiological signal, progesterone, 2) the natural oocyte arrests in oocyte maturation at either prophase I (equivalent to a late G2-arrest) or at MII (M-phase arrest), 3) and the large size of the oocyte (1.2 mm in diameter) and its high protein content (30 mg soluble proteins per cell), offering the rare opportunity of single-cell proteomics of events at the single cell level. In addition, the *Xenopus* model offers great assets. From a technical perspective, in the mouse, the small number of oocytes and their low protein content limit proteomic approaches (30,000 oocytes isolated from 950 mice were required for the phosphoproteomic profiling by (Sun *et al*, 2024)). Furthermore, the resumption of meiotic divisions in mouse obeys a particular regulation: unlike other mammalian models (including the human species) and other vertebrates whose entry into meiosis I depends strictly on protein translation, it does not require the synthesis of new proteins, making it a somewhat marginal

model (Meneau *et al*, 2020 [link](#)). For these reasons, we performed time-resolved proteomics and phosphoproteomics in *Xenopus* oocytes from prophase I through the MII arrest of the unfertilized egg.

In this study, we have analyzed the changes in the phosphorylation in relation to three categories of events: specific periods of meiotic maturation, activities of the master regulators of meiosis and peculiar cellular features of meiosis. Our analysis confirms the high quality of our phosphoproteome by the detection of multiple phosphorylation events with previously-documented functional effects on meiotic divisions or early development. Through the detection of many new regulated phosphosites, it provides a rich pool of candidate proteins for multiple avenues of investigation of previously uncharacterized important players of the oocyte to embryo transition.

Results

The temporal resolution of the events of meiotic divisions

Prophase-arrested oocytes (PRO) and oocytes resuming meiosis were collected at regular intervals after progesterone exposure. To correlate the time-points of this experiment with the cytological events known to occur during meiosis resumption, a set of proteins whose accumulation had been extensively studied during *Xenopus* oocyte meiosis progression were used as markers (**Figure 2A** [link](#)-video supplement 1). One of the first events following progesterone stimulation is the accumulation of Cyclin B1 that begins before and independently of Cdk1 activation (Santoni *et al*, 2024 [link](#); Frank-Vaillant *et al*, 1999 [link](#)). In our experiments, Cyclin B1 accumulation starts at about 2 hours and increases linearly until MII arrest at 6.5 hours (**Figure 2A** [link](#)). It has been reported that Mos translation is stimulated by progesterone concomitant with the accumulation of Cyclin B1; however, the Mos protein does not accumulate before NEBD because it turns over rapidly (Santoni *et al*, 2024 [link](#); Frank-Vaillant *et al*, 1999 [link](#)). Another critical well documented event that occurs downstream to Cdk1 activation is the degradation of CPEB1 (Santoni *et al*, 2024 [link](#); Mendez *et al*, 2002 [link](#)), an RNA-binding protein whose degradation activates protein translation in MI. Hence, Mos accumulation and the degradation of CPEB1 are well established markers for the timing of NEBD. After NEBD, activation of the anaphase-promoting complex (APC) leads to the ubiquitination and degradation of Cyclin B2, which marks the entry into anaphase I. During the MI-MII transition, multiple additional events take place: Cyclin B3 is degraded (Bouftas *et al*, 2022 [link](#)), Cdc6 (Lemaître *et al*, 2002 [link](#); Whitmire *et al*, 2002 [link](#); Daldello *et al*, 2015 [link](#)), Wee2 (Nakajo *et al*, 2000 [link](#)) and Cyclin E (Rempel *et al*, 1995 [link](#)) accumulate, reaching their maximal levels in MII. These four events mark the entry into MII. Based on these markers, the timing of cytological events in our experiment is as follows (**Figure 2A** [link](#)): NEBD and MI occur between 3 and 4 hours, the MI-MII transition takes place between 5 and 6 hours, and entry and arrest in MII are at 7 hours. This time course is in agreement with the cytological previously described events (Huchon *et al*, 1981 [link](#); Gard, 1992 [link](#)). A physiological measure of MII arrest can be ascertained by the ability of the oocytes to be activated by an electric shock, a property that is acquired only in MII (**Figure 2** [link](#)-video supplement 2).

Validation of the quantitative proteomic data

The quality of oocytes resuming meiosis is also confirmed by the dynamics of protein accumulation and degradation that occur during meiotic progression (**Figure 2B** [link](#)). A set of 34 proteins, whose changes in concentration have been previously studied during meiotic divisions was used to validate the quality of our quantitative proteomics data (**Figure 2B** [link](#)-figure supplement 3A). Our proteome correctly classifies 15 out of 18 proteins accumulating during *Xenopus* meiosis resumption ($FC(MII/PRO) > 1.5$), 11 out of 12 proteins whose concentration is constant ($0.75 < FC(MII/PRO) < 1.5$), and 4 out of 4 proteins whose accumulation was reported to

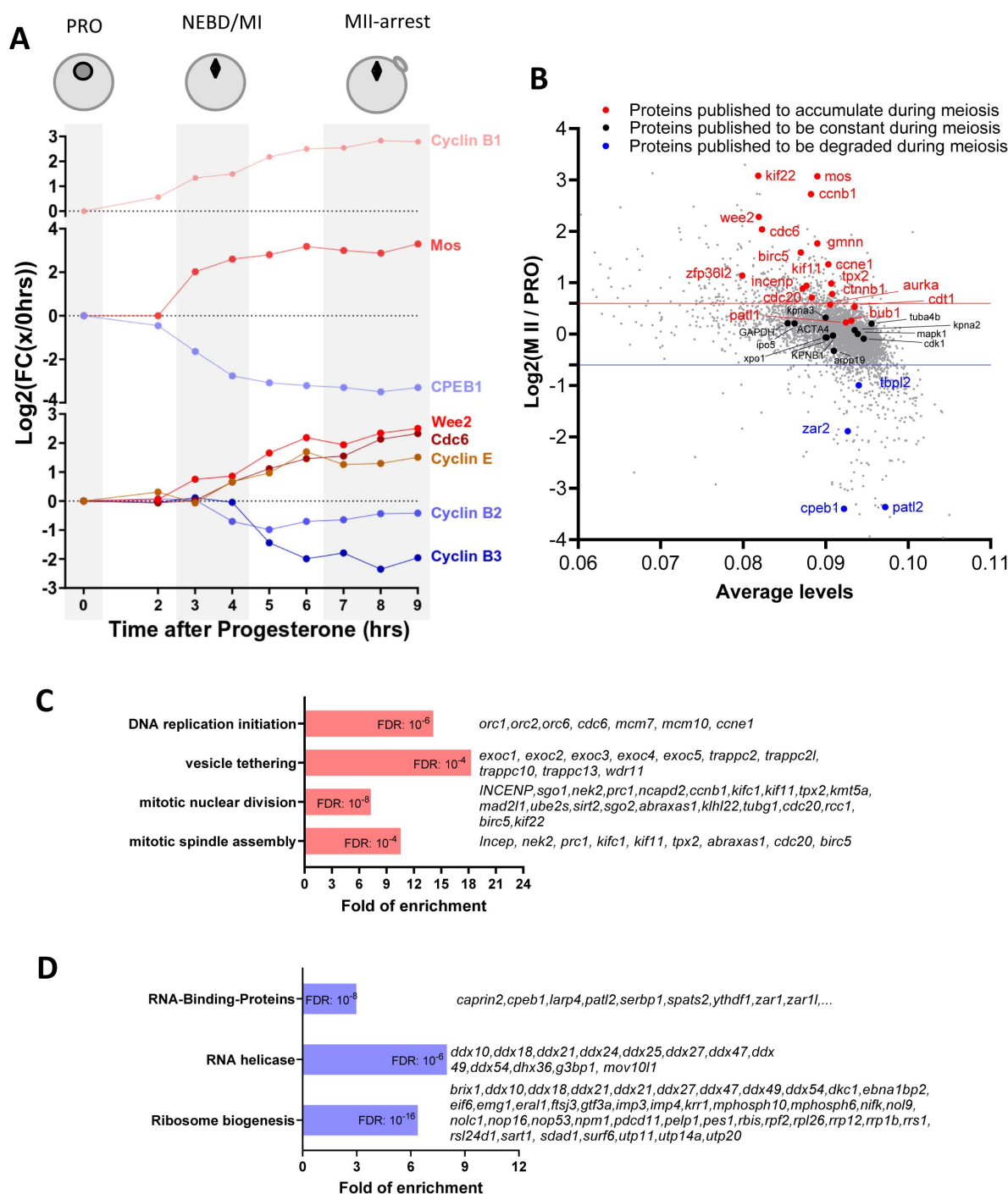


Figure 2.

Changes in the protein levels during meiotic divisions. **(A)** The relative changes in the concentration of proteins whose accumulation/degradation has been reported to correlate with the critical stages of *Xenopus* meiotic divisions have been plotted. **(B)** Changes of protein stoichiometry during meiotic maturation. The log2 fold changes between the concentration of each protein in PRO-arrested oocytes and MII-arrested oocytes was plotted. The red line marks the 1.5-fold change used as a threshold to identify accumulating proteins. The blue line marks the 0.75-fold change used as threshold to distinguish proteins that are degraded during meiosis. Proteins, whose changes in homeostasis during meiotic divisions are documented in the literature, are marked with the following color code: red, for proteins accumulating; black, for proteins which are expressed at a stable level; and blue, for proteins that are degraded. **(C-D)** Gene ontology analysis of the biological processes enriched among the proteins found to accumulate (in red, panel **C**) or degraded (in blue, panel **D**) during meiosis. The statistical significance of the enrichment is expressed as false discovery rate (FDR). A representative subset of the proteins belonging to each group is displayed.

decrease ($FC(MII/PRO) < 0.75$) (**Figure 2B** [figure supplement 3A](#)). The concordance between our dataset (30/34 proteins) with the changes in the protein levels reported in the literature is substantially higher than in a previously published system-wide proteomic dataset from *Xenopus* oocytes where only 7 out of these 34 proteins were correctly measured (Peuchen *et al*, 2017 [figure supplement 3A-B](#)). This also may explain why changes in protein concentration reported in that study poorly correlate with the one measured in our proteomic study (**Figure 2** [figure supplement 3B](#)). The high quality of the new proteomic data is also confirmed by the strong correlation of the translational pattern of *Xenopus* allo-alleles (**Figure 2** [figure supplement 3C-D](#)), which were derived by genome endoduplication (Session *et al*, 2016 [figure supplement 3C-D](#)).

Protein homeostasis during meiotic divisions

Absolute concentrations of proteins during meiotic maturation

The absolute concentration of 7974 proteins identified in our proteome dataset was calculated (See Appendix 1). We then performed a qualitative comparison of average oocyte abundance of proteins belonging to distinct functional groups (see Appendix 2). As expected, the less abundant subgroups include signaling molecules and transcription factors. Perhaps of more surprise, E3 enzymes and kinases are not abundant. At the other end of the spectrum are glycolytic and tricarboxylic acid cycle enzymes, as well as proteasome and ribosome components. These observations are expected because the oocyte has a stock of nutrient molecules and all the enzymes linked to their metabolism, which are used in viviparous animals throughout embryonic and larval development. For instance, the glycolytic enzyme GAPDH that catalyzes an important energy-yielding step in carbohydrate metabolism, is a very abundant protein in the oocyte (average concentration of between 8.88 μ M and 17.83 μ M). The ribogenesis program, which takes place during the oocyte growth from stage III to stage VI, before meiosis resumption, allows the accumulation of very large quantities of ribosomal RNAs and proteins that are used during embryonic development, until the swimming tadpole stage (Woodland, 1974 [figure supplement 3E](#)). Similarly, the components of the DNA replication machinery are quite abundant, probably in anticipation of the post-fertilization cell cycles. All these data are therefore consistent with biological understanding of their roles during embryonic development after post-fertilization (Peshkin *et al*, 2015 [figure supplement 3E](#)).

Temporal profiles of protein levels

We clustered the relative data for protein abundance across subsequent hourly timepoints post-progesterone stimulation (see Methods). As illustrated in Appendix 3A, the majority of the 60 expression profiles are flat, demonstrating that most (over 80%) proteins are stable during meiotic maturation. Our proteomic dataset allowed to identify 12% of proteins affected by changes in concentration during meiotic progression by applying the same fold change threshold: 413 proteins that accumulate and 562 proteins whose concentration decreases (**Figure 2B** [figure supplement 3E](#); see Appendix 4). Although these changes only concern a minority of proteins, they are critical for the meiotic process. Since transcription is silent during this period, these documented changes must result from the regulation of translation or/and degradation (Meneau *et al*, 2020 [figure supplement 3E](#)).

Protein accumulation: the machinery of cell division and DNA replication

Gene ontology analysis was used to characterize the functional significance of the 413 proteins whose level increases during meiosis. Proteins involved in spindle assembly and mitotic division were significantly enriched (**Figure 2C** [figure supplement 3F](#)). Among these important regulators are the Targeting-protein-for-Xklp2 (Tpx2) and Protein-regulator-of-cytokinesis 1 (Prc1) that also accumulate during meiotic maturation in mouse oocytes (Brunet *et al*, 2008 [figure supplement 3F](#); Li *et al*, 2021 [figure supplement 3F](#)). We also find that the centrosomal maturation factor, SSX2IP, and two plus-end motors, the kinesins Kifc1 and Kif11 (Eg5) (Houliston *et al*, 1994 [figure supplement 3F](#)), also accumulate significantly during meiotic maturation. Three of

these proteins, Tpx2 and the two kinesins, play an important role in the assembly of both acentrosomal meiotic spindles and centrosomal mitotic spindles (Gruss, 2018 [↗](#); Kufer *et al*, 2002 [↗](#); Ems- McClung *et al*, 2020; Miller *et al*, 2019 [↗](#)). Interestingly, the prophase oocyte is equipped with most of the proteins essential for the formation of division spindles. Hence these new results suggest that increases in the relative level of some components is required to support the meiotic and first rapid embryonic divisions.

Another protein category found to increase is the machinery of DNA replication initiation (**Figure 2C** [↗](#)). Indeed, oocytes accumulate the components of the DNA replication machinery during meiosis resumption to support the 13 rapid rounds of cell cycles occurring during early embryogenesis, all of which occur in the absence of transcription. In *Xenopus*, it was shown that Cdc6 is the only missing factor of the pre-replicative (pre-RC) complex in the prophase-arrested oocyte and that its accumulation occurs during the MI-MII transition (Lemaître *et al*, 2002 [↗](#); Whitmire *et al*, 2002 [↗](#); Daldello *et al*, 2015 [↗](#)) (**Figure 1** [↗](#)). Translational and then post-translational mechanisms targeting Cdc6 prevent the pre-RC from functioning between the two meiotic divisions, thus ensuring the production of a haploid gamete. Our proteomic analysis shows that, even though the other pre-RC proteins, as ORC1, ORC2 and ORC6, are already expressed in prophase oocytes, they accumulate further during meiotic maturation, revealing a strong co-regulation of the components of the pre-RC.

Another process controlled by highly enriched proteins is vesicle tethering, which includes the family of exocyst proteins (**Figure 2C** [↗](#)). Interestingly, exocyst proteins are involved in the completion of cell division during the secretory-vesicle-mediated abscission (Gromley *et al*, 2005 [↗](#)). The accumulation of these proteins during meiotic maturation could be important to support the extrusion of the polar bodies, as already suggested in *C. elegans* (Kumar *et al*, 2019 [↗](#)) and the rapid cell divisions sustaining early embryogenesis.

Protein degradation: RNA-binding proteins (RBPs)

Gene ontology analysis of the 562 proteins whose concentration decreases during meiosis reveals that 116 (21 %) are annotated to bind RNA. Among them, a large portion are RNA helicases (**Figure 2D** [↗](#)). Among these RBPs, another important group are proteins that are involved in ribosome biogenesis. The ribogenesis program takes place during the oocyte growth, allowing the accumulation of ribosomal RNAs and proteins that are used during embryonic development (Woodland, 1974 [↗](#)). This process is downregulated during meiotic maturation (Hyman & Wormington, 1988 [↗](#); Chen *et al*, 2011 [↗](#)). The decrease in the concentration of proteins involved in rRNA synthesis and processing does not result in a general decrease of ribosomal proteins that are stable during meiotic divisions (**Figure 2** [↗](#)-figure supplement 4A). Halting ribogenesis may allow the oocyte to re-allocate its energy resources to the production of other cellular components required for meiotic divisions and embryogenesis. Intriguingly, Rpl26 (Ribosomal protein L26) is the only ribosomal protein detected to decrease during meiotic divisions (**Figure 2** [↗](#)-figure supplement 4A). Since this protein is known to be dispensable for ribosome formation in yeast (Babiano *et al*, 2012 [↗](#); Warner & McIntosh, 2009 [↗](#); Takagi *et al*, 2005 [↗](#)), the distinct regulation of Rpl26 as compared to the other Rpls suggests that Rpl26 might have extra- ribosomal functions. This hypothesis is reinforced by its specific modification by the ubiquitin-like UFMylation modification (Walczak *et al*, 2019 [↗](#)) and its role in the regulation of protein stability (Zhang *et al*, 2016 [↗](#)). Another subgroup of RBPs that decrease during meiotic divisions is involved in the repression of translation during prophase arrest, such as CPEB1 (Mendez *et al*, 2002 [↗](#); Reverte *et al*, 2001 [↗](#); Piqué *et al*, 2008 [↗](#)), PATL2 (Nakamura *et al*, 2010 [↗](#); Cao *et al*, 2021 [↗](#); Christou-Kent *et al*, 2018) and Zar11 (Zar2) (Charlesworth *et al*, 2012 [↗](#); Yamamoto *et al*, 2013 [↗](#); Rong *et al*, 2019 [↗](#)). Remodeling the repertoire of RBPs in the oocyte could be a general mechanism to regulate translation during meiotic divisions. Among these RBPs is Serbp1, which in mouse is involved in the stabilization of mRNA involved in meiosis resumption (Chew *et al*, 2013 [↗](#)), as well as Caprin2, whose degradation is also observed during meiotic maturation of human oocytes (Virant-Klun *et al*,

2016). Caprin2 is an RBP with unknown function and highly enriched in the oocyte Balbiani body (Boke *et al*, 2016), a non-membrane compartment specific to the early diplotene oocytes, which contains mitochondria, RNAs and endoplasmic reticulum. The functions of this organelle are not clear, although it is strongly correlated in *Xenopus* to the establishment of oocyte polarity and to the localization of maternal determinants (Jamieson-Lucy & Mullins, 2019). In *Xenopus*, it disperses at the onset of vitellogenesis, leaving a wedge-shaped cytoplasmic region rich in mitochondria (Wilk *et al*, 2005) and promotes the formation of RNP condensates in the vegetal cortex of the oocyte (Yang *et al*, 2022). Interestingly, Rbpms2, another regulator of the Balbiani body (Kaufman *et al*, 2018), is strongly degraded during meiotic maturation, highlighting how dynamic during meiosis is the composition of this transient compartment specific to the oocyte.

Interestingly, distinct E3 ubiquitin ligases, mainly APC and SCF (Skp, Cullin, F-box containing complex) have been shown to be active at different time periods of meiosis resumption in *Xenopus* oocytes (Kinterová *et al*, 2022). 186 proteins (37%) identified as decreasing in our proteome include in their sequences one of the APC degrons (Figure 2-figure supplement 4B). This group of proteins includes two known APC targets, Cyclin B2 (Glötzer *et al*, 1991) and Securin (Cohen-Fix *et al*, 1996). Additionally, 30 proteins (6%) whose concentration decreases in the oocytes have one SCF degron in their sequence (Figure 2-figure supplement 4B), including CPEB1 (Reverte *et al*, 2001) and Cdc6 (Daldello *et al*, 2015), two characterized SCF-targets. Interestingly, 253 proteins (57%) of proteins that decrease during meiotic divisions do not bear in their sequence any of the degrons for the ubiquitination systems identified in *Xenopus* oocytes (Figure 2-figure supplement 4B). Further research is therefore required to identify the sequences and the E3 enzymes and deubiquitinating enzymes that control the turnover of this important protein class.

The highly dynamic landscape of phosphorylation during meiotic maturation

We performed phosphoproteomic mass spectrometry analysis in order to measure the overall dynamics of protein phosphorylation taking place during meiotic divisions and to identify which specific proteins and which sites are phosphorylated during this process. We identified 6783 different phosphopeptides and quantified the dynamics of relative change across all time points (see Appendix 5). These peptides came from 2308 distinct proteins. When respective protein and un-phosphorylated peptide were also measured, we were able to compute the stoichiometry (see Methods). A list of all 415 phospho-sites for which it was possible to calculate the phospho-occupancy is provided as (see Appendix 6).

We identified the human homologous counterparts of 5901 (87%) among the 6783 sites found to be phosphorylated during meiotic divisions. Interestingly, only 177 (3%) out of 5901 conserved sites were already known to be phosphorylated on *phosphosite.org* database. Thus, our phosphosite dataset identifies a very large number of novel phosphosites, never previously identified and characterized. Such a small number of previously reported phosphorylations among those detected by our phosphoproteome might suggest that a large portion of these phosphorylation events are related to cell division. Dividing cells generally account for a small fraction of tissue mass. This leads to a high dilution of the phosphorylation signature of cell division, as compared to the highly synchronous meiotic divisions of oocyte maturation. Interestingly, 60% of the phosphosites detected in oocytes are dynamically regulated during meiotic maturation, highlighting the importance of this post-translational regulation in controlling this last step of oogenesis and meiosis. Unbiased clustering reveals that changes in protein phosphorylation are pervasive and far more dynamic than changes in protein abundance (see Appendix 3), which was also noted during yeast meiosis (Koch *et al*, 2024). Comparing the three stages of meiotic division (PRO, MI and MII) and based on the main cellular events of these stages, we can group phosphopeptide stoichiometry into 5 classes (Figure 3A; see Appendix 5). Class I includes proteins bearing a phosphosite whose occupancy decreases during the first 2 hours following progesterone stimulation. This class includes only 136 phosphopeptides (2%) which might be

involved in the early signaling pathway induced by progesterone and PKA downregulation. This small number was expected since there is a low level of phosphorylation detectable in prophase-arrested oocytes (Maller *et al*, 1977 [\[1\]](#)). This makes sense since there are presumably only a few substrates of PKA sufficient to initiate meiosis resumption. Class II is larger (39% of the phosphopeptides) and includes proteins whose phosphorylation increases between NEBD and MI and which remain highly phosphorylated in both MI and MII. The massive amount of Class II phosphosites presumably reflects the catalytic power of Cdk1 as well as its downstream kinases, to generate the thousands of substrates essential for cell division. Our measurements that most of the sites, which are phosphorylated in MI, remain stably phosphorylated during MI to MII transition is consistent with the observations made in yeast showing that most of Cdk1 motifs remain phosphorylated at the end of meiosis I (Celebic *et al*, 2024 [\[2\]](#)). Class III (5%) consists of proteins whose phosphorylation peaks in MI then decreases while oocytes progress throughout meiotic maturation. This is the most predominant class of phosphosites present during meiotic maturation in starfish (Swartz *et al*, 2021 [\[3\]](#)) and follow the peculiar pattern of Cdk1 activity peaking at MI in these oocytes (Okano-Uchida *et al*, 1998 [\[4\]](#); Kishimoto, 2003 [\[5\]](#)). Class IV (9%) is composed of proteins whose phosphorylation progressively increases from MI to MII, hence potentially implicated in MII entry. Finally, Class V (5%) comprises proteins whose phosphorylation increases from MII and which could play a role in establishing the MII arrest. The phosphosites that are specific for either MI (Class III) or MII (Classes IV and V) are much less abundant (5%, 9% and 5% respectively, **Figure 3A** [\[6\]](#)) and might distinguish the biochemical state of the two meiotic divisions.

The release of the prophase arrest: receptors and membrane trafficking

Receptors

Since meiotic maturation signaling begins with the interaction of progesterone with membrane receptors, we first focused our analysis on these proteins (Josefsberg Ben-Yehoshua *et al*, 2007; Sadler & Maller, 1985 [\[7\]](#)). Progesterone signals through activation of a seven-pass-transmembrane progesterone receptor (mPR β or PAQR8) that belongs to the progestin and adiponectin receptor family (Josefsberg Ben-Yehoshua *et al*, 2007; Nader *et al*, 2018 [\[8\]](#), 2020 [\[9\]](#)) (**Figure 1** [\[10\]](#)). The release of the prophase block also involves the inactivation of an orphan constitutively active GPCR, GPR185, responsible for maintaining oocyte prophase arrest by ensuring high cAMP levels and PKA activity (Ríos-Cardona *et al*, 2008 [\[11\]](#); Deng *et al*, 2008 [\[12\]](#)). Progesterone decreases GPR185 signaling either through its cleavage by a metalloproteinase (Deng *et al*, 2008 [\[12\]](#)) or through its endocytosis (Nader *et al*, 2014 [\[13\]](#)). One limitation of our dataset lies in the loss of some membrane proteins during the fractionation protocol, explaining why our proteomic workflow did not detect mPR β and GPR185 or its close relative GPR12. Nevertheless, it does detect some proteins that interact with mPR, *i.e.* PGRMC1, VLDLR and APPL1. PGRMC1 is known to form a receptor complex with mPR α and is necessary for mediating progesterone signaling in zebrafish oocytes (Wu *et al*, 2018 [\[14\]](#)). Interestingly, our phosphorylation data reveals that PGRMC1 and its close relative PGRMC2 are phosphorylated on two homologous threonine residues during meiotic maturation, suggesting that these proteins may contribute to mPR β activity in *Xenopus* oocytes. In the prophase-arrested oocyte, VLDLR regulates mPR trafficking from the endoplasmic reticulum through the Golgi to the plasma membrane, preparing the oocyte to be responsive to progesterone (Nader *et al*, 2018 [\[8\]](#)). In contrast, progesterone induces clathrin-dependent endocytosis of mPR β into signaling endosomes, where mPR interacts transiently with APPL1 and Akt2 to induce meiosis (Nader *et al*, 2020 [\[9\]](#)). The phosphoproteomic data reveal that both APPL1 and Akt are phosphorylated in response to progesterone at the time of Cdk1 activation, suggesting that phosphorylation events regulate the traffic and activity of this receptor.

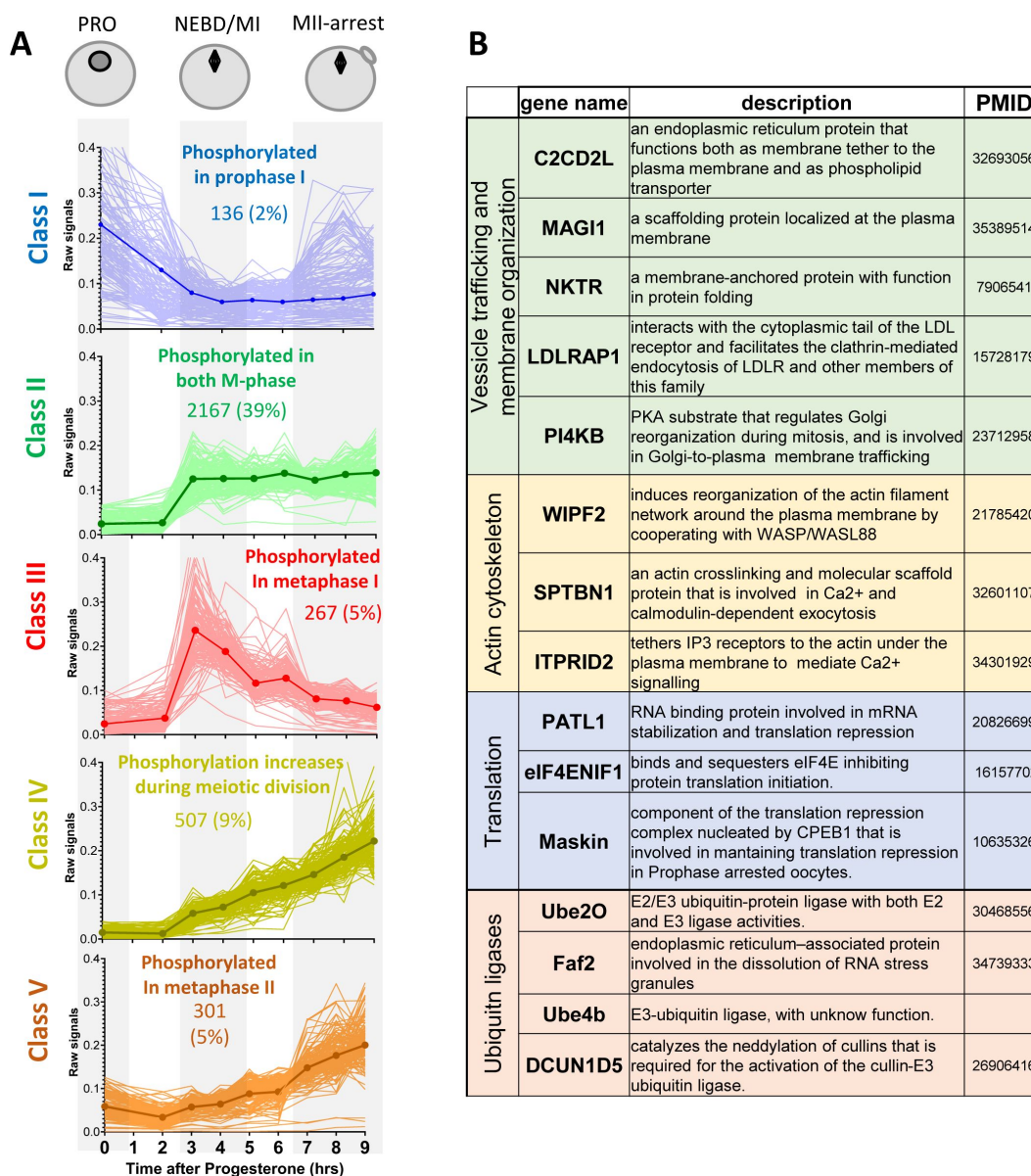


Figure 3.

Changes in protein phosphorylation during meiotic divisions. **(A)** Relative phosphopeptide signals were normalized for the changes in the total level of the proteins. The changes in the phosphopeptides were measured between the three biological stages of meiotic divisions (PRO = 0hr, NEBD/MI = average of 3 and 4 hours, and MII-arrest = average of 7 to 9 hours). Class I (n=136 - 2%, in blue) includes proteins bearing a phosphosite whose occupancy decreases during meiosis resumption ($\text{Log2FC}(\text{MI}/\text{PRO}) < -1$). Class II (n=2167 - 39%, in green) contains proteins that are highly phosphorylated from MI ($\text{Log2FC}(\text{MI}/\text{PRO}) > 1$) to MII ($-1 < \text{Log2FC}(\text{MII}/\text{MI}) < 1$). Class III (n=267 - 5%, in red) consists of proteins whose phosphorylation is specific of MI ($\text{Log2FC}(\text{MI}/\text{PRO}) > 1$, $\text{Log2FC}(\text{MII}/\text{MI}) < -1$). Class IV (n=507 - 9%, yellow) is composed of proteins whose phosphorylation increases from MI ($\text{Log2FC}(\text{MI}/\text{PRO}) > 1$) to MII, picking in MII ($\text{Log2FC}(\text{MII}/\text{MI}) > 1$). Class V (n=301 - 5%, orange) comprises proteins whose phosphorylation does not vary in MI ($-1 < \text{Log2FC}(\text{MI}/\text{PRO}) < 1$) but are phosphorylated only in MII ($\text{Log2FC}(\text{MII}/\text{MI}) > 1$). The average signal of each Class was plotted with a thick and dark line. **(B)** Table summarizing the proteins bearing phosphosites belonging to Class I.

Membrane traffic

Vesicular trafficking at the cell membrane appears to be crucial for the maintenance and the release the prophase meiotic arrest, and may also function through early embryogenesis (**Figure 1**). In the prophase-arrested oocyte, the accumulation of the GPR185 receptor at the plasma membrane maintains high cAMP levels (Nader *et al*, 2014) while the plasma membrane targeting of mPR β renders the oocyte competent to respond to progesterone (Nader *et al*, 2018).

Exocytosis is also crucial for the formation of the fluid-filled blastocoele cavity during embryogenesis (Müller, 2001). Indeed, the apical membrane of the epithelium surrounding the blastula, where the polarized activities of ion channels and transporters generate the blastocoele fluid, is formed from the oocyte cell membrane (Müller, 2001). This requires further remodeling. Progesterone rapidly blocks this exocytosis pathway (Colman *et al*, 1985; Leaf *et al*, 1990), leading to a decrease in membrane surface area, which is revealed by the disappearance of microvilli that are enriched in oocytes but almost absent in eggs (Dumont, 1972; Kado *et al*, 1981; Larabell & Chandler, 1989; Bluemink *et al*, 1983). Importantly, blocking exocytosis induces meiotic maturation, in the absence of hormonal stimulation (Mulner-Lorillon *et al*, 1995; El-Jouni *et al*, 2007). Furthermore, progesterone stimulates endocytosis of membrane proteins, such as GPR185, whose activity is suppressed by its internalization, and mPR β , whose internalization is required to transduce its effects (Nader *et al*, 2018, 2020, 2014). The stimulation of endocytosis also converts plasma membrane into intracellular vesicles that can provide membrane reserves, necessary to support the rapid cell divisions, and hence the requirement for greater total membrane surface area, during embryogenesis (Angres *et al*, 1991; Gawantka *et al*, 1992). However, it is unknown whether PKA is responsible for the active secretory transport during the prophase arrest and how progesterone blocks exocytosis and stimulates endocytosis (**Figure 1**).

To improve our understanding of vesicular trafficking and membrane organization, we analyzed the proteins involved in these processes whose phosphosites belong to Class I (**Figure 3B**). Among these proteins, PI4KB deserves special attention. This protein is critical for the maintenance of the Golgi and trans-Golgi phosphatidylinositol-4-phosphate (PI4) pools. PI4KB regulates Golgi disintegration/reorganization during mitosis and is involved in Golgi-to-plasma membrane trafficking (De Matteis *et al*, 2013). The phosphoproteome reveals that PI4KB is dephosphorylated within 2 hours after progesterone stimulation, on a PKA consensus phosphorylation site conserved in vertebrates. The phosphorylation of this site by PKA is known to control the interaction of PI4KB with Armadillo-like helical domain-containing protein 3 (ARMH3), an interaction important for the Golgi membrane integrity (McPhail *et al*, 2020; Isobe *et al*, 2017; Blomen *et al*, 2015). PI4KB is also regulated by its interaction with ACBD3, an AKAP-like scaffold platform in Golgi, which directly binds the R regulatory subunit of PKA and regulates the traffic between Golgi and endoplasmic reticulum in a PKA-depending manner (Jia *et al*, 2023; Klima *et al*, 2016; Sasaki *et al*, 2012). The phosphoproteome reveals that ACBD3 is phosphorylated during meiotic maturation at the time of NEBD (Class II). PI4KB is therefore at the crossroad between PKA and intra-membrane endoplasmic reticulum-Golgi-plasma membrane trafficking. Since these are key events involved in meiosis resumption, PI4KB is a particularly attractive candidate as a PKA substrate for future functional studies of membrane relocation.

Proteins involved in the actin cytoskeleton are also represented in Class I phosphosites (**Figure 3B**). Their early post-translational modifications could regulate the reorganization of the actin cytoskeleton that is known to accompany the modification of the secretory/endocytosis pathways (Gard, 1999). Among them, ITPRID2/KRAP tethers IP3 receptors, which are located in the membrane of the endoplasmic reticulum, to the actin under the plasma membrane, to mediate Ca²⁺ signalling (Thillaiappan *et al*, 2021). During meiotic maturation, the endoplasmic reticulum, which is the major Ca²⁺ store, is enriched in the cortex of the oocyte (Campanella *et al*, 1984;

Charbonneau & Grey, 1984 [\[1\]](#); Terasaki *et al.*, 2001 [\[2\]](#)). This remodeling brings the Ca^{2+} source close to its primary targets at fertilization, when Ca^{2+} release from endoplasmic reticulum activates the oocyte and blocks polyspermy. The proteins highlighted above (**Figure 3B** [\[3\]](#)) could participate to this critical reorganization of the endoplasmic reticulum.

Protein translation and accumulation orchestrate meiotic divisions

Two waves of translation take place in the oocytes: the first depends on PKA inactivation. It occurs upstream and independently of Cdk1 activity and is required for Cdk1 activation. The second translation wave takes place downstream Cdk1 activation (Santoni *et al.*, 2024 [\[4\]](#)). Hence, in contrast to mitosis, a period when protein synthesis is repressed (Ross, 1997 [\[5\]](#)), translation is activated during meiotic maturation, indicating the existence of meiosis-specific controls of the process. However, the regulation of the two waves of protein translation and the identity of the newly translated proteins had not been unraveled. Interestingly, protein translation components are extensively regulated at the level of phosphorylation as demonstrated by the enrichment of the proteins regulating these processes among 3 of the 5 phosphopeptide classes (**Figure 4A** [\[6\]](#)).

RBPs and translational control

Interestingly, several proteins implicated in the control of translation bear phosphosites belonging to Class I, as eIF4ENIF1/4E-T, PATL1 and TACC3/Maskin (**Figs. 3B** [\[7\]](#) and **4B** [\[8\]](#)). eIF4ENIF1/4E-T is required for eIF4e1b localization to P-bodies where mRNAs are stored in a dormant state in zebrafish oocytes (Lorenzo-Orts *et al.*, 2023 [\[9\]](#)). Both 4E-T and eIF4e1b are extensively phosphorylated on multiple residues during meiotic divisions, suggesting a highly dynamic regulation of P-bodies during this process (**Figure 4B** [\[10\]](#)). Many RBPs known to control translation in oocytes are also regulated at the phosphorylation level, including Pum proteins (Nakahata *et al.*, 2003 [\[11\]](#); Ota *et al.*, 2011 [\[12\]](#); Padmanabhan, 2006 [\[13\]](#)), Zar proteins (Charlesworth *et al.*, 2012 [\[14\]](#); Yamamoto *et al.*, 2013 [\[15\]](#); Rong *et al.*, 2019 [\[16\]](#); Heim *et al.*, 2022 [\[17\]](#)) and PATL proteins (Christou-Kent *et al.*, 2018; Zhang *et al.*, 2023 [\[18\]](#); Marnef *et al.*, 2010 [\[19\]](#)). PATL1 is a RNA-binding protein required for cytoplasmic mRNA P-body assembly in oocytes (Marnef *et al.*, 2010 [\[19\]](#)). PATL1 and PATL2 have mutually exclusive expression patterns in *Xenopus* oocytes: PATL2 is degraded while PATL1 accumulates during meiotic maturation (Marnef *et al.*, 2010 [\[19\]](#)), as confirmed by our proteomic data (**Figure 2B,D** [\[20\]](#)-figure supplement 3). Although the role of PATL2 and the regulation of its stability by phosphorylation have been well documented in the oocyte (Christou-Kent *et al.*, 2018; Zhang *et al.*, 2023 [\[18\]](#)), little is known about the role of PATL1 during meiosis. Both PATL1 and 2 inhibit translation when tethered to mRNA and assemble a complex that includes CPEB1, Xp54/DDX6, Rap55B/LSM14b (Marnef *et al.*, 2010 [\[19\]](#)) (**Figure 4B** [\[21\]](#)). The early dephosphorylation of PATL1 detected in the phosphoproteome could modulate this inhibitory complex during early events of meiotic maturation, while the extensive phosphorylation of the other components of the complex could control its activity later during meiotic divisions (**Figure 4B** [\[22\]](#)). Another component of the repressive complex that interacts with CPEB1 is TACC3/Maskin, which is found widely regulated during meiotic divisions in the phosphoproteome (**Figure 4B** [\[23\]](#)). TACC3/Maskin plays a dual function, regulating the centrosome-mediated microtubule nucleation through γ -TuRC (Peset *et al.*, 2005 [\[24\]](#)) but also translation (Barnard *et al.*, 2005 [\[25\]](#); Groisman *et al.*, 2002 [\[26\]](#)). Eight residues in TACC3/Maskin, of which only two are conserved in human Maskin (214/228 and 298/317 in human/*Xenopus*), exhibit a complex profile of phosphorylation (**Figure 4B** [\[27\]](#)). Two sites identified in the phosphoproteome, S266 and S298, were previously found to be phosphorylated in egg extracts, regulating the function of TACC3/Maskin in the control of translation (Barnard *et al.*, 2005 [\[25\]](#)). S626 was also reported to be phosphorylated in prophase oocytes and during oocyte meiotic maturation, although there is some controversy about the kinase (either PKA or Aurora-A) and the role of this phosphorylation in either the control of translation or centrosome attachment (Barnard *et al.*, 2005 [\[25\]](#); Pascreau *et al.*, 2005 [\[28\]](#)). Interestingly, in our studies S240 is phosphorylated in the prophase oocyte and dephosphorylated early in response to progesterone (Class I) (**Figure**

4B [↗](#)). Since the sequence surrounding S240 does not correspond to a PKA consensus motif, it is hard to establish the biological significance; nevertheless, such an early dephosphorylation deserves further attention.

The control of translation initiation

Translation regulation might not only depend on the expression and post-translational regulation of the RBPs but could also involve the regulation of the core translation initiation machinery, since we find that many of its components are phosphorylated (**Figure 4C** [↗](#)). Indeed, eIF4b, an activator of the eIF4a helicase that facilitates ribosome scanning ([Jackson et al, 2010](#) [↗](#)), eIF2b/eIF2s2, a component of the molecular machinery recruitment on the met-tRNA ([Jackson et al, 2010](#) [↗](#)), and eIF4g, involved in the recognition of the mRNA 5'Cap ([Jackson et al, 2010](#) [↗](#)), are among the proteins bearing sites whose phosphorylation peaks in MII (**Figure 4** [↗](#)-figure supplement 1). These phosphorylation patterns correlate with the second wave of activation of translation that begins under the control of Cdk1 activity in MI and continues progressively through MII ([Meneau et al, 2020](#) [↗](#); [Santoni et al, 2024](#) [↗](#)).

The control of ubiquitin ligases

The levels of specific proteins are also regulated by proteolysis at each step of meiotic maturation. Targeted regulation allows for the accumulation of key proteins independently of any general increase in their translation, such as for Cyclin B1 ([Santoni et al, 2024](#) [↗](#)), or the well-studied degradation of proteins, such as Cyclin B2, securin or CPEB1 ([Taleb et al, 2001](#) [↗](#); [Wassmann, 2022](#) [↗](#); [Setoyama et al, 2007](#) [↗](#)). Interestingly, proteins controlling polyubiquitination display phosphopeptides enriched in Class II, strengthening the view that a major regulation of protein turnover takes place at NEBD in order to orchestrate meiotic divisions (**Figures 3** [↗](#) and **5A** [↗](#)). APC is the key E3 ubiquitin ligase that promotes metaphase-anaphase transition. Although the phosphorylation status of some substrates determines their recognition by APC and influences the precise timing of their degradation, APC activity is also regulated by the phosphorylation of its core subunits ([Bansal & Tiwari, 2019](#) [↗](#)) (**Figure 5B** [↗](#)).

Another important ubiquitination system that regulates stability of proteins during mitosis and meiosis is the SCF complex ([Kinterová et al, 2022](#) [↗](#)). SCF is composed of three main components: Cul1, a scaffold protein; Skp1, which interacts with specific F-box proteins that are involved in the substrate recognition; and Rbx1, which interacts with the E2-ubiquitin ligases ([Kinterová et al, 2022](#) [↗](#)) (**Figure 5C** [↗](#)). SCF ligases play important roles during oocyte meiotic maturation, especially the SCF^{βTrCP} complex. SCF^{βTrCP}-mediated degradation of Emi1, Emi2/Erp1, CPEB1 and BTG4 is essential for progression through MI and MII of mouse and *Xenopus* oocytes ([Kinterová et al, 2022](#) [↗](#)). This is very similar to the role of Emi1 in mitosis and is thought to set the time between binding of cyclin B and its destruction ([Reimann et al, 2001](#) [↗](#)).

Many F-Box proteins are also expressed in *Xenopus* oocytes, but they do not display high phosphorylation dynamics. An exception is Lmo7/Fbxo20, which is degraded during meiosis, and is dynamically phosphorylated (**Figure 5C** [↗](#)). Interestingly, Lmo7 overexpression causes defects at the spindle assembly checkpoint, affecting the progression through mitotic divisions ([Tzeng et al, 2018](#) [↗](#)).

Additionally, many E2 enzymes were identified in our datasets as highly regulated at the translational/accumulation and post-translational levels during meiotic maturation (**Figure 5C** [↗](#)). This layer of regulation of protein ubiquitination remains understudied. Interestingly, an E2/E3 hybrid ubiquitin- protein ligase, UBE2O, UBE4B (also known as UBOX3 or E4) and FAF2 (or Ubxn3B/UBXD8) display phosphorylation sites belonging to Class I (**Figures 3B** [↗](#) and **5C** [↗](#)). FAF2 is involved in stress granule clearance ([Gwon et al, 2021](#) [↗](#)). UBE2O is also known to be involved in the drastic proteome remodeling during erythroid differentiation ([Nguyen et al, 2017](#) [↗](#)), as well as quality control factor for orphans of multiprotein complexes ([Yanagitani et al, 2017](#) [↗](#)).

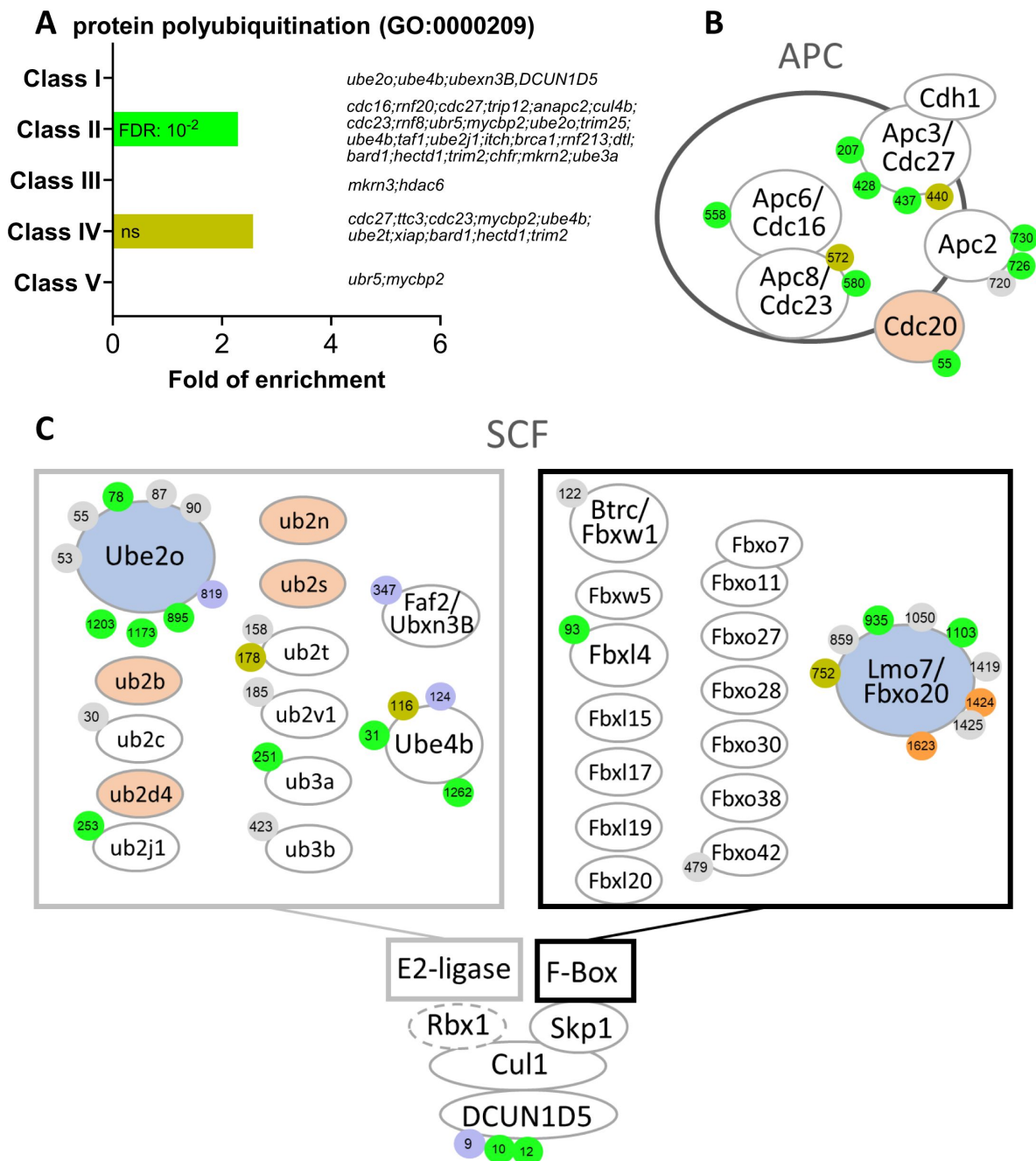


Figure 5.

E2 and E3 ubiquitin ligases are highly regulated by phosphorylation. **(A)** Gene ontology enrichment analysis of the biological process "protein polyubiquitination" (GO:0000209) among the phosphopeptides found in each class defined in **Fig. 3A**. **(B-C)** Graphical representation of the phosphorylation dynamics of the components of two E3 ubiquitin ligases, the anaphase promoting complex (APC) **(B)** and the Skp, Cullin, F-box containing complex (SCF) **(C)** during meiotic divisions. Proteins are color-coded to highlight changes in protein abundance during meiotic divisions: unchanged=white, decreased=blue, increased=light orange. The phosphorylated sites are color-coded: unchanged, grey; Class I, blue; Class II, green; Class III, red; Class IV, yellow; Class V, orange. The numbering is referred to the iso-allele which has higher number of phosphorylation sites detected.

Ribosomal proteins are known substrates of UBE2O which leads to their degradation (Nguyen *et al*, 2017 [\[1\]](#)). Whether UBE2O is involved in the downregulation of ribosome biogenesis that occurs during meiotic maturation (Hyman & Wormington, 1988 [\[2\]](#)), and in the decrease of Rpl26 (Figure 2-figure supplement 4A), is an interesting question. The dephosphorylation of UBE2O, UBE4B and FAF2 could also be involved in the accumulation of Cyclin B1, which was recently shown to occur independently of any increase in translation but to result from protein stabilization (Santoni *et al*, 2024 [\[3\]](#)).

Another important regulator of the SCF system, DCUN1D5, is found regulated in our phosphoproteome. DCUN1D5 catalyzes the neddylation of all cullins, which is necessary for the activation of cullin-RING E3 ubiquitin ligases (Keuss *et al*, 2016 [\[4\]](#)). DCUN1D5 is dephosphorylated at S9, a PKA consensus site, and phosphorylated starting NEBD at S10 and S12 (Figure 5C [\[5\]](#)). These phosphorylations could control DCUN1D5 ability to activate SCF, especially since the inhibition of neddylation causes a meiotic arrest in MI in mouse oocytes (Yang *et al*, 2019 [\[6\]](#)). Altogether, our dataset provides attractive candidates to be involved in the regulation of both translation and protein stability that occur during meiotic maturation.

Activation of Cdk1 occurs through an intricate network of phosphorylation of Cdk1 regulators

The activation of Cdk1 during oocyte maturation relies on an intricate network of feedforward and feedback phosphorylation pathways mediated by kinases and phosphatases (Figure 1 [\[7\]](#)). Indeed, components of the cell division machinery are highly enriched among Class II, III, IV and V (Figure 6A [\[8\]](#)). Some of the phosphorylation sites implicated in cell division control have been identified previously through detailed studies using site-specific mutagenesis and phospho-specific antibodies. Our phosphoproteomic analysis of oocyte maturation provides new regulatory elements (Figure 6B [\[9\]](#)). An important regulator of the Cdk1 auto-amplification loop is the kinase Plk1 that contributes to the phosphorylation of the two direct Cdk1 regulators, Cdc25 and Myt1 (Myt1, a membrane-bound protein, was not detected in our proteome). Plk1 is activated by Aurora-A, a kinase itself under the indirect control of Cdk1 in *Xenopus* oocyte (Maton *et al*, 2003 [\[10\]](#)), through the phosphorylation of T210 (T201 in *Xenopus*) (Macůrek *et al*, 2008 [\[11\]](#)), a residue localized in the activation T-loop of the kinase domain. We show here that Plk1 phosphorylation at T210 increases in MI and is then constant throughout meiotic maturation (Class II) (Figure 6B-figure supplement 1A). This is consistent with the function of Plk1, which phosphorylates multiple proteins required for the mechanics of division, and which must logically be kept active throughout both meiotic divisions (Hansen *et al*, 2006 [\[12\]](#); Solc *et al*, 2015 [\[13\]](#)). Interestingly, the phosphoproteome also detects additional phosphorylation sites of Plk1, S326 (S335 in human) and S340 (not conserved in human), which were previously found to be upregulated in response to okadaic acid (Wind *et al*, 2002 [\[14\]](#)), an inhibitor of PP2A that strongly induces M-phase. These two residues are located between the two functional domains of Plk1, the kinase domain and the Polo-box domain (Cheng, 2003 [\[15\]](#)). This suggests that the phosphorylation of other critical residues than T210 of Plk1 could be essential for its catalytic activation. Aurora-A kinase is activated during meiotic maturation downstream Cdk1 (Maton *et al*, 2003 [\[10\]](#); Castro *et al*, 2003 [\[16\]](#); Maton *et al*, 2005 [\[17\]](#)). While our phosphoproteome does not detect Aurora-A autophosphorylation at T288 (T295 in *Xenopus*), it registers the extensive phosphorylation of Bora, an activator of Aurora-A involved in Plk1 activation (Thomas *et al*, 2016 [\[18\]](#)) (Figure 6B-figure supplement 1B). Several phosphorylation sites of Bora have been reported to be important for its function as an Aurora-A activator: S41/38, S112/110, S137/135 and S252/S252 (human/*Xenopus*) (Thomas *et al*, 2016 [\[18\]](#); Vigneron *et al*, 2018 [\[19\]](#); Tavernier *et al*, 2015 [\[20\]](#), 2021 [\[21\]](#)). Among these sites, S110 phosphorylation of Bora by Cdk1-Cyclin A is critical for mitotic entry in *Xenopus* egg extracts (Vigneron *et al*, 2018 [\[19\]](#); Tavernier *et al*, 2015 [\[20\]](#), 2021 [\[21\]](#)). Our phosphoproteome reveals that Bora belongs to Class II, being phosphorylated starting NEBD, but surprisingly on distinct sites that were never reported in the literature: S191/189, T294/287, S552/549, S-521, T12/9 and T15/12 (human/*Xenopus*), with the exception of S278/285. In contrast to the situation in mitosis, Cyclin A is very weakly expressed in

the prophase oocyte and accumulates during meiotic maturation (Kobayashi *et al*, 1991 [\[1\]](#)). Our proteome dataset reveals that Cyclin A accumulates after MI. Hence, the phosphorylation of Bora at MI cannot be catalyzed by Cdk1-Cyclin A in the *Xenopus* oocyte. Cdk1-Cyclin B is probably involved in these phosphorylations that take place concomitantly with its activity and at Cdk1 phosphorylation consensus sites for 5 of the 7 phosphosites (**Figure 6B** [\[1\]](#)-figure supplement 1B). Thus, this analysis reveals that Bora activation correlates with its phosphorylation at residues not previously identified in earlier studies of the oocyte. We also detect additional phosphorylation events on other players of the Cdk1 amplification loop, as Cdc25 and Gwl/MASTL (**Figure 6B** [\[1\]](#)-figure supplement 1C- D), whose extensive phosphorylation during M-phase is usually detected by a large electrophoretic shift (Dupré *et al*, 2014 [\[2\]](#)).

Phosphoregulation of the bimodal activity of Cdk1 during MI-MII transition

In *Xenopus* oocytes, Cdk1 is not fully inactivated at anaphase I and during extrusion of the 1st polar body. Indeed, the low Cdk1 activity that persists during the MI-MII transition is essential to avoid the reformation of a nucleus, chromosome decondensation and DNA replication (Huchon *et al*, 1993 [\[3\]](#); Thibier *et al*, 1997 [\[4\]](#)). This short period is followed by a further increase of Cdk1 activity, due to a sustained synthesis of B-cyclins; the increased Cdk1 activity allows the formation of the MII spindle (Gerhart *et al*, 1984 [\[5\]](#)) (**Figure 1** [\[1\]](#)). Although this bimodal subtle regulation of Cdk1 activity is essential for the success of meiosis, *i.e.* linking two successive divisions without intervening DNA replication, it has never been quantified precisely. Cdk1 activity can be measured by looking at the phosphorylation of one of its direct substrates, Cdc27, a core APC component, at S428 (S426 in human) (Patra & Dunphy, 1998 [\[6\]](#)). Our precise quantification of the phosphorylation occupancy of this site shows that its phosphorylation is virtually absent in prophase, increases by 40% in MI, decreases during the MI-MII transition to 20%, then increases again in MII by 60% (**Figure 6** [\[1\]](#)-figure supplement 1E). Three other residues of Cdc27, T205/207, S435/437 and S438/440 (human/*Xenopus*), also known to be phosphorylated at M-phase (Kraft, 2003 [\[7\]](#)), follow the same bimodal pattern than S428. The subsequent increase in Cdk1 activity essential for MII entry requires that the kinase escapes inhibitory phosphorylations by Myt1 and Wee2. Both Myt1 and Wee2 are reported to be inhibited by phosphorylations, although the sites responsible for this inhibition are not precisely defined. Myt1, a membrane protein, was not detected in our proteome. In *Xenopus*, Wee2 is not expressed in prophase oocytes and accumulates after NEBD (Nakajo *et al*, 2000 [\[8\]](#); Charlesworth *et al*, 2012 [\[9\]](#)) (**Figure 1** [\[1\]](#)). Our analysis reveals that newly synthesized Wee2 is immediately phosphorylated at 4 sites (Classes II and IV), one of them being not conserved in human, and none of them corresponding to the proposed inhibitory S53 and S123 sites (**Figure 6A-B** [\[1\]](#)-figure supplement 1F). Hence, these data provide exciting new avenues of research to discover unexpected new regulations of old players in Cdk1 activation.

The Mos/MAPK pathway activation

Mos plays a well-known role at the origin of a critical pathway controlling meiotic maturation (Yew *et al*, 1992 [\[10\]](#); Furuno *et al*, 1994 [\[11\]](#)). It phosphorylates Mek1/2, which in turn phosphorylates Erk1/2 (also known as MAPK) which leads to Rsk1/2 activation (**Figure 6B** [\[1\]](#)). These play multiple essential roles, such as preventing the reformation of a replicating nucleus between MI and MII (Dupré *et al*, 2002 [\[12\]](#)), enabling arrest in MII (Sagata *et al*, 1989b [\[13\]](#)), but also participating in Cdk1 activation (Sagata *et al*, 1989a [\[14\]](#)) and the functioning of meiotic spindles (Araki *et al*, 1996 [\[15\]](#); Choi *et al*, 1996 [\[16\]](#); Bodart *et al*, 2005 [\[17\]](#)). The phosphoproteome detects the activation of the Mos-MAPK module, with the phosphorylation of Erk2 on T185 and Y187 (T188 and Y190 in *Xenopus*) by MEK1/2 (Payne *et al*, 1991 [\[18\]](#)), an event that begins in MI (**Figure 6B** [\[1\]](#)-figure supplement 2A). Interestingly, additional phosphorylation sites are identified in MEK1 and Rsk1/2, the latter being hyperphosphorylated during meiosis as detected by a large electrophoretic shift (Bhatt & Ferrell, 2000 [\[19\]](#)) (**Figure 6B** [\[1\]](#)-figure supplement 2B-E). Rsk1 auto-phosphorylation at S381 (S380 in human) is required for its activity (Vik & Ryder, 1997 [\[20\]](#)). Our phosphoproteome measures a highly-

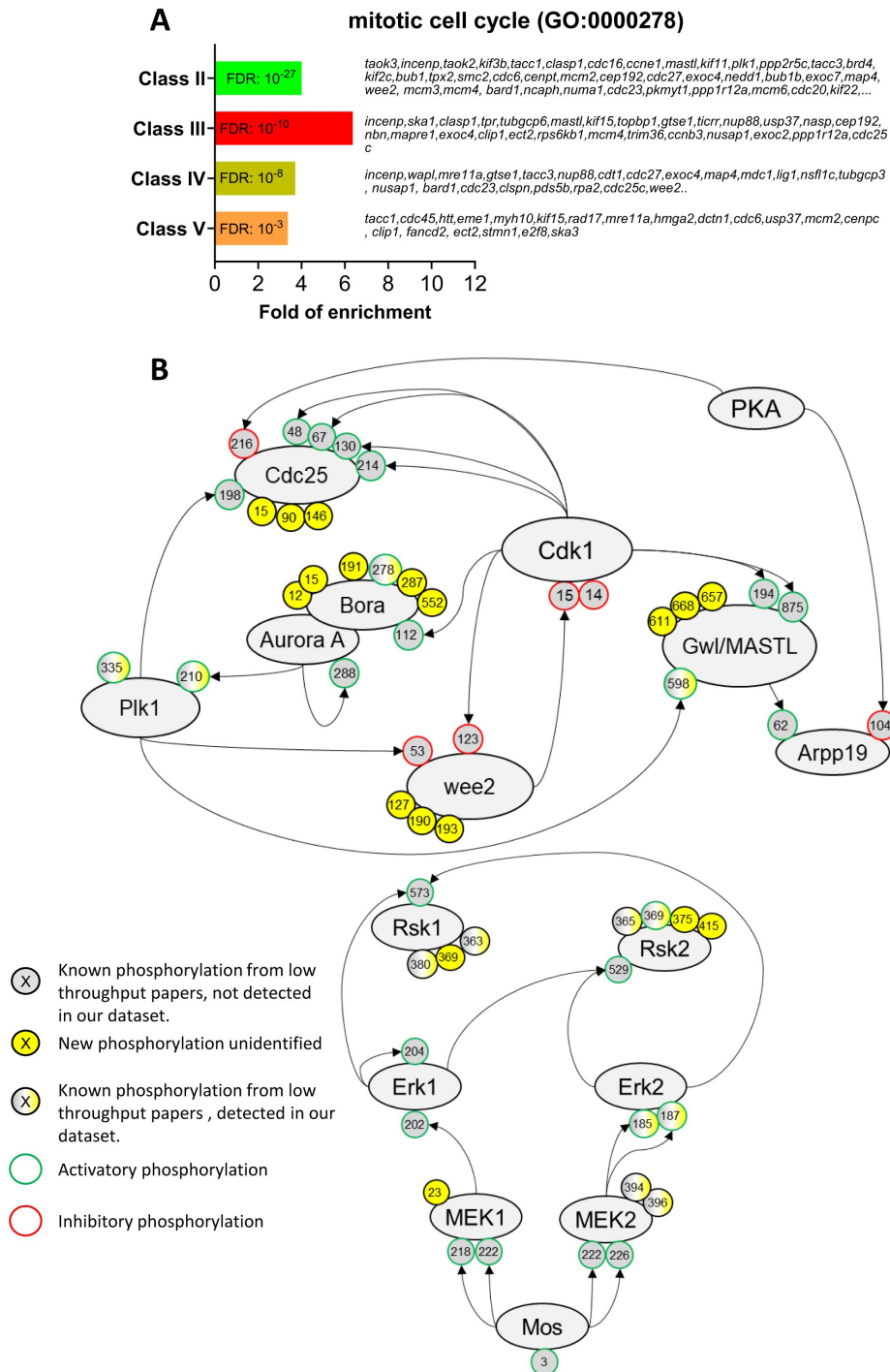


Figure 6.

The network of phosphorylations regulating Cdk1 and the Mos/MAPK pathway. **(A)** Gene ontology enrichment analysis of the biological process “mitotic cell cycle” (GO:0000278) among the phosphopeptides found in each class defined in **Fig. 3A**. **(B)** Graphical representation of the network of phosphorylations of cell cycle regulators controlling Cdk1 activation (**upper panel**) and the Mos/MAPK pathway (**lower panel**). The human nomenclature numbering has been used for the phosphosites. The phosphosites are displayed by dots and color-coded: yellow background, new sites identified in this paper and never described elsewhere; grey background, phosphosites already documented in vertebrates by low throughput papers and not identified in this paper; grey-yellow background, phosphosites already documented in vertebrates by low throughput papers and also identified in this paper. The functional effect of the phosphorylation is indicated by the color of the dot contour line: green, activatory; red, inhibitory; black, unknown.

confident phospho-occupancy for this site demonstrating the switch-like activation of the MAPK pathway at NEBD, followed by the maintenance of a constant level of its activity (**Figure 6B** [figure supplement 2F](#)). The timing of activation of the Mos-MAPK cascade has been controversial, one hypothesis being that it is activated before Cdk1 and participates in the activation of this kinase ([Sagata *et al*, 1988](#) [Gavin *et al*, 1999](#)), while another is that it is under the control of the starter Cdk1 activity and is only involved in the autoamplification loop ([Santoni *et al*, 2024](#) [Dupre *et al*, 2002](#) [Fisher *et al*, 1999](#)). The phosphoproteomic data clearly support the latter.

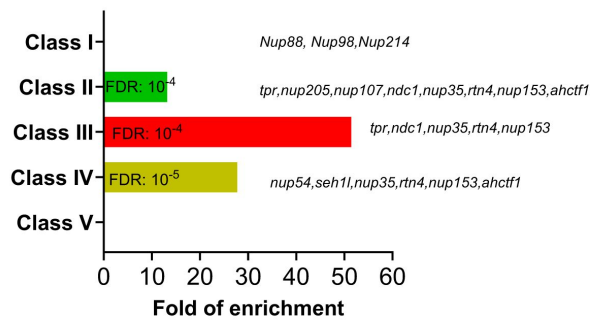
Phosphorylation of key components of the nuclear envelope during breakdown

As a result of the massive activation of kinase activities under the control of Cdk1, hundreds of proteins are phosphorylated and orchestrate the structural events choreographing the mechanics of meiotic divisions. The first of these is the massive vesiculation of the nuclear envelope surrounding the enormous oocyte nucleus. During mitosis, nuclear pore proteins are extensively phosphorylated by Cdk1-Cyclin B, Plk1 and Nek6/7 to promote the breakdown of the nuclear envelope (Kutay *et al.* 2021 [1](#)). The *Xenopus* oocyte has the unusual feature of having an enormous nucleus (475 μm in diameter), and therefore a considerable surface area of the nuclear envelope (0.96 mm^2), as well as large stocks of annulate lamellae in the cytoplasm. Annulate lamellae are cytoplasmic stack cisternae of nuclear envelope perforated with nuclear pores, 10 times more abundant than in the nucleus in the *Xenopus* oocyte, but devoid of lamins (Bement & Capco, 1990 [2](#); Miller & Forbes, 2000 [3](#); Cordes *et al.*, 1995 [4](#)). Their surface area also far exceeds that of the nuclear envelope surrounding the nucleus (Cordes *et al.*, 1995 [4](#)). This correlates with the high protein concentrations of the nuclear pore proteins (0.1 to 0.7 μM) that we have measured (**Figure 7A** [5](#)). Annulate lamellae play a critical role as a reservoir for the formation of the 4000 of nuclei that form during the embryonic cell divisions leading up to the mid-blastula transition and the onset of transcription (Kessel *et al.*, 1986 [6](#); De Magistris & Antonin, 2018 [7](#); Feldherr, 1974 [8](#)). During meiotic maturation, NEBD and vesiculation of annulate lamellae occur at the same time, in a progressive spatial manner, starting from the vegetal pole. Importantly, the nucleus and annulate lamellae do not re-form between the two meiotic divisions, reducing the risk that an S-phase can take place between MI and MII (Bement & Capco, 1990 [2](#)). A gene ontology analysis of our phosphoproteome highlights that the phosphorylation of nuclear pore components is enriched in Classes II, III and IV, displaying the highest level of enrichment in Class III, hence characteristic of MI (**Figure 7B** [5](#)). The nucleoporin Nup53 (also called Nup35) is a component of the soluble core region of the nuclear pore complex that is extensively phosphorylated in mitosis by Cdk1 and Plk1 (Linder *et al.* 2017 [9](#)). Phosphomimetic mutations of all Cdk1 and Plk1 sites in Nup53 slow down the reformation of nuclear pores in interphase and mutation to phospho-null amino acids impairs the nuclear breakdown during meiosis (Linder *et al.* 2017 [9](#)). Our phosphoproteome reveals that Nup53 phosphorylation is more dynamic than previously believed (**Figure 7C** [5](#)). Indeed, 6 phosphopeptides are found to peak in MI (Class III), 9 phosphopeptides are phosphorylated both in MI and MII (Class II), and 2 phosphopeptides continuously increase in phosphorylation during meiotic maturation (Class IV) (**Figure 7C** [5](#)). We were able to calculate the phospho-occupancy of the S59 (S66 in human), showing that it increases from 30% in prophase to 90% in MI oocytes and stays constant until MII (Class II) (**Figure 7D** [5](#)). Several other transmembrane nucleoporins, notably NDC1 and POM121C, undergo dramatic increase of phosphorylation at transition to MI. Altogether, these observations indicate that dissociation of the nuclear envelope and the annulate lamellae occurring in MI might require either a specific set of phosphorylations and/or a general level of phosphorylation that is higher than the level of phosphorylation required to maintain the nucleus and the annulate lamellae dissociated during the MI- MII transition through MII.

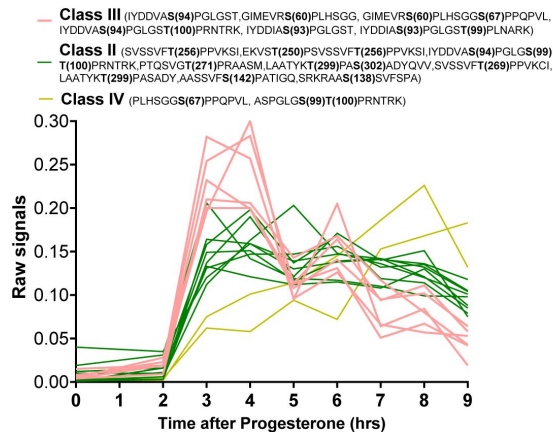
A Nucleoporins concentration (nM)

	PRO	MII
NUP107	157	161
NUP133	166	172
NUP153	117	144
NUP155	186	191
NUP160	134	140
NUP188	120	124
NUP205	360	383
NUP210	97	119
NUP214	150	131
NUP35	327	308
NUP37	309	317
NUP43	218	237
NUP50	712	725
NUP54	160	166
NUP58	385	388
NUP62	440	440
NUP85	134	145
NUP88	110	110
NUP93	665	693
NUP98	476	442

B Nuclear pore organization (GO:0006999)



C Nup35/Nup53



D Nup35/Nup53 GVMEMRSPLFSGG Xen(59) / Hum(59)

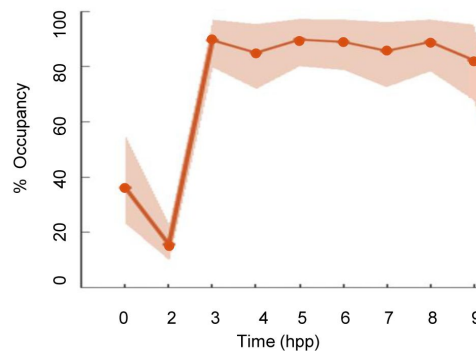


Figure 7.

The phosphorylation pattern of nuclear pore components during meiotic divisions. (A) Absolute concentration (nM) of nucleoporins in prophase (PRO) and metaphase II (MII) oocytes. (B) Gene ontology enrichment analysis of the biological process “nuclear pore organization” (GO:0006999) among the phosphopeptides found in each class defined in [Fig. 3A](#). (C) Phosphorylation patterns of Nup35/Nup53. The phosphorylation curves are color-coded: Class II, green; Class III, red; Class IV, yellow. (D) High precision phospho-occupancy calculation of S59 phosphorylation pattern of Nup35/Nup53.

The regulation of centrosomal proteins during oocyte maturation

An intriguing feature of oocytes in most, if not all, metazoan species is the elimination of their centrioles. This allows the sperm to contribute its centriole and thereby initiates the division cycle and the cell cycle. There is longstanding literature that proves that the oocyte or egg has retained its ability to generate centrioles. However, in nature, the female centrioles are lost and the sperm centriole is the main contributor. In *Xenopus*, oocytes entering prophase I contain a typical centrosome, including a pair of centrioles and pericentriolar material (PCM) containing γ -tubulin and pericentrin. Centrioles disappear at the end of the pachytene stage of prophase I and PCM disperses in multiple foci in the cytoplasm that are no longer functional as microtubule nucleation sites (Gard *et al*, 1995 [\[1\]](#)). At fertilization, the male gamete delivers two centrioles but does not provide the PCM (Delattre & Gönczy, 2004 [\[2\]](#)). Therefore, in most species (with the exception of rodents), the complete centrosome of the one-cell stage embryo is a bi-parental inheritance, reconstituted from paternal centrioles and maternal PCM (Delattre & Gönczy, 2004 [\[2\]](#)). In *Xenopus*, oocytes store enough quiescent centrosomal building blocks for duplicating and assembling 1000 to 2000 centrosomes (Gard *et al*, 1990 [\[3\]](#)). This implies that oocytes should comprise an enormous reserve of centrosomal proteins, *i.e.* centriolar and PCM proteins, that sustain the reconstitution of a functional centrosome at fertilization and its subsequent rounds of duplication in the early embryo.

Centriolar proteins

The centriole is formed by a polarized cylinder of microtubule triplets, decorated along the proximal to distal axis with several appendages that define the three main regions of the organelle: the proximal part, the central core, and the distal region (LeGuennec *et al*, 2021 [\[4\]](#)). In agreement with a published RNA-seq and proteome of *Xenopus* eggs (Session *et al*, 2016 [\[5\]](#); Wühr *et al*, 2014 [\[6\]](#)), our proteome reveals that most of the centriolar proteins are expressed in the oocyte (Table 1 [\[7\]](#)). The components of the proximal part, as SAS-6 and CPAP/CENPJ, and of the central core, as POC1B, POC5 and Centrin 1, are either constant or accumulate during meiotic maturation (Table 1 [\[7\]](#)). In contrast, proteins of distal and subdistal appendages are present at lower concentration, either constant or decreasing during meiotic maturation (Table 1 [\[7\]](#)). Indeed, the appendages are critical for the docking of mature centrioles at the plasma membrane and ciliogenesis (Tanos *et al*, 2013 [\[8\]](#)), a process that is not functional in the oocyte and during the early embryonic divisions (Drysdale & Elinson, 1992 [\[9\]](#)).

Besides the basic building blocks forming the centriole, the oocyte is expected to provide the machinery for its duplication, the first cycle occurring immediately after fertilization. The key regulators of centriole duplication, including Plk4, SAS-6 and STIL, are generally expressed at very low levels (Bauer *et al*, 2016 [\[10\]](#)). While Plk4 is expressed at the mRNA level in oocytes but escaped our proteomic detection, SAS-6 and STIL are expressed at a constant level during meiotic maturation (Table 1 [\[7\]](#)). The estimated concentrations of both proteins are in agreement with the previous predictions, that cartwheels comprise one molecule of STIL for every SAS-6 dimer (Bauer *et al*, 2016 [\[10\]](#)). Other important regulators of centriole duplication, such as CEP192, CEP152, CEP63, RTTN, CEP110/CCP110, CEP97 and CEP76 were also detected, and are either constant or accumulate during meiotic divisions (Table 1 [\[7\]](#)). Interestingly, our phosphoproteome reveals that several of these proteins are phosphorylated during meiotic maturation. STIL is phosphorylated at S554, which is located in a Cdk1 phosphorylation consensus (Table 1 [\[7\]](#)-table supplement 1A), an interesting observation since phosphorylation of STIL by Cdk1-Cyclin reduces the efficiency of daughter centriole assembly. This phosphorylation provides an explanation why centrioles cannot duplicate during M-phase (Zitouni *et al*, 2016 [\[11\]](#); Steinacker *et al*, 2022 [\[12\]](#)). A massive CEP97 phosphorylation occurs at MI at 8 distinct sites, only 3 of them lying within a Cdk1 consensus motif (Table 1 [\[7\]](#)-table supplement 1B). Remarkably, one of the phosphosites of CEP97, S633, has been recently shown to be phosphorylated by the kinase Dyrk1a in multiciliated cells of *Xenopus*

	Gene name	Proteome										(Whur,2014)	(Session,2016)
		0	2	3	4	5	6	7	8	9	Log2 FC (MII/PRO)	Estimated concentration in MII-arrested oocytes (nM)	Count per Million (CPM) in PRO I
Proximal part	SAS-6	0.11	0.09	0.10	0.10	0.09	0.09	0.09	0.09	0.08	-0.3	36	20
	CENPJ	0.07	0.09	0.09	0.10	0.10	0.11	0.09	0.10	0.09	0.4	9	56
	Cep135	not detected										5	1
Central core	POC1B	0.05	0.05	0.07	0.08	0.09	0.14	0.11	0.10	0.11	1.2	7	96
	POC5	0.05	0.06	0.07	0.08	0.09	0.10	0.10	0.11	0.12	1.1	39	100
	CETN1	0.08	0.08	0.11	0.09	0.09	0.09	0.10	0.09	0.10	0.2	not detected	3
	SFI1	not detected										3	7
	POC1A	not detected										20	54
	CETN2	not detected										82	not detected
	WDR90	not detected										5	25
	FAM161A	not detected										not detected	34
Distal appendages	SCLT1	0.10	0.09	0.10	0.11	0.10	0.10	0.08	0.08	0.08	-0.4	13	2
	FBF1	0.11	0.12	0.11	0.12	0.11	0.11	0.08	0.07	0.06	-0.7	not detected	17
	CEP83	not detected										not detected	7
	CEP89	not detected										not detected	1
Sub-distal appendages	FBF1	0.11	0.12	0.11	0.12	0.11	0.11	0.08	0.07	0.06	-0.7	not detected	17
	CEP164	0.10	0.08	0.11	0.09	0.12	0.09	0.08	0.06	0.08	-0.4	1	not detected
	NIN	0.11	0.09	0.11	0.07	0.08	0.11	0.10	0.08	0.10	-0.2	4	27
	TUBE1	0.10	0.09	0.10	0.10	0.10	0.11	0.09	0.09	0.09	-0.1	7	8
	TUBD1	0.08	0.08	0.08	0.09	0.09	0.11	0.09	0.09	0.10	0.1	13	28
	ODF2	not detected										7	8
	CEP128	not detected										3	17
	CNTRL	not detected										6	26
	CEP170	not detected										4	3
	TCHP	not detected										2	16
	CCDC68	not detected										not detected	3
	tubz1	not detected										not detected	10
	CCDC120	not detected										not detected	not detected
Centriole duplication	CCP110	0.06	0.07	0.06	0.08	0.08	0.09	0.10	0.12	0.12	0.9	7	20
	CEP152	0.07	0.07	0.10	0.10	0.10	0.12	0.10	0.11	0.09	0.6	7	161
	CEP192	0.08	0.08	0.08	0.08	0.09	0.09	0.10	0.11	0.10	0.5	21	24
	STIL	0.08	0.08	0.07	0.08	0.10	0.11	0.10	0.11	0.12	0.5	14	100
	CEP63	0.07	0.06	0.11	0.11	0.13	0.12	0.10	0.09	0.09	0.5	4	33
	CEP76	0.07	0.07	0.07	0.08	0.09	0.14	0.10	0.07	0.12	0.5	12	57
	RTTN	0.09	0.09	0.09	0.10	0.10	0.11	0.09	0.09	0.08	0.0	18	14
	CEP97	0.10	0.10	0.10	0.10	0.09	0.10	0.10	0.09	0.09	-0.1	15	1
	SAS-6	0.11	0.09	0.10	0.10	0.09	0.09	0.09	0.09	0.08	-0.3	36	42
	Plk4	not detected										not detected	not detected
PCM	KIF22	0.02	0.02	0.04	0.05	0.07	0.09	0.12	0.14	0.19	3.1	33	464
	SSX2IP	0.03	0.03	0.05	0.07	0.09	0.11	0.11	0.13	0.16	2.1	20	10
	TUBG1	0.04	0.04	0.04	0.04	0.04	0.04	0.12	0.12	0.11	1.5	166	101
	tpx2	0.06	0.06	0.07	0.08	0.10	0.11	0.11	0.12	0.11	1.0	43	516
	kif11	0.06	0.06	0.07	0.08	0.09	0.10	0.10	0.11	0.12	0.9	44	185
	dync11i1	0.06	0.0735	0.0605	0.0675	0.062	0.0605	0.104	0.1185	0.087	0.8	169	38
	dync2h1	0.067	0.057	0.073	0.074	0.092	0.098	0.104	0.105	0.128	0.7	35	3
	prc1	0.06	0.07	0.08	0.09	0.10	0.12	0.09	0.10	0.10	0.7	32	13
	CEP152	0.07	0.07	0.10	0.10	0.10	0.12	0.10	0.11	0.09	0.6	7	33
	kif4a	0.08	0.07	0.08	0.09	0.09	0.09	0.10	0.10	0.10	0.4	100	120
	TUBGCP4	0.08	0.08	0.08	0.09	0.10	0.08	0.10	0.10	0.10	0.2	14	20
	nedd1	0.08	0.08	0.09	0.08	0.09	0.08	0.10	0.10	0.09	0.2	30	12
	dync21i	0.0875	0.076	0.0915	0.088	0.0965	0.0975	0.0925	0.0935	0.101	0.1	not detected	11
	CKAP5	0.09	0.09	0.09	0.09	0.09	0.10	0.09	0.10	0.09	0.1	67	102
	dync112	0.086	0.111	0.077	0.102	0.09	0.08	0.086	0.085	0.104	0.1	411	60
	dync212	0.089	0.081	0.1	0.099	0.106	0.098	0.09	0.087	0.095	0.0	not detected	35
	TUBGCP3	0.09	0.08	0.09	0.09	0.10	0.10	0.09	0.10	0.09	0.0	33	10
	KATNB1	0.09	0.09	0.10	0.10	0.10	0.10	0.09	0.09	0.09	0.0	20	108
	dync11	0.086	0.1	0.114	0.09	0.09	0.07	0.083	0.085	0.093	0.0	138	49
	dync112	0.091	0.104	0.104	0.073	0.081	0.106	0.091	0.089	0.092	0.0	232	48
	PCNT	0.09	0.09	0.09	0.10	0.10	0.10	0.09	0.09	0.09	0.0	16	23
	NPM2	0.08	0.10	0.08	0.12	0.08	0.08	0.07	0.07	0.09	0.0	1181	116
	dync21i1	0.086	0.074	0.084	0.086	0.089	0.091	0.082	0.081	0.091	0.0	43	4
	CLASP1	0.09	0.08	0.09	0.10	0.10	0.10	0.09	0.09	0.08	-0.1	15	48
	TUBGCP2	0.09	0.09	0.10	0.10	0.10	0.09	0.09	0.09	0.09	-0.1	59	22
	dync11i2	0.097	0.089	0.099	0.104	0.094	0.099	0.091	0.094	0.086	-0.1	26	27
	dync11	0.099	0.101	0.094	0.096	0.092	0.105	0.089	0.093	0.089	-0.1	51	130
	NEK9	0.09	0.10	0.10	0.10	0.10	0.10	0.09	0.08	0.08	-0.2	45	27
	TACC3	0.10	0.10	0.10	0.10	0.10	0.10	0.09	0.09	0.08	-0.2	115	117
	TUBGCP6	0.10	0.09	0.11	0.10	0.10	0.10	0.08	0.08	0.08	-0.3	8	14
	NPM3	0.11	0.09	0.12	0.11	0.10	0.10	0.09	0.09	0.08	-0.3	150	70
	KATNA1	0.10	0.10	0.10	0.09	0.10	0.09	0.09	0.08	0.08	-0.3	63	30
	NPM1	0.12	0.09	0.12	0.10	0.09	0.09	0.08	0.08	0.08	-0.6	308	102
	TUBGCP5	not detected										24	not detected
	KATNAL1	not detected										5	not detected
	KATNAL2	not detected										not detected	22

Table 1.

Centrosomal components during meiotic divisions.

Expression patterns of the components of the centrioles and the pericentriolar material. The changes in protein abundance are color-coded: unchanged=white, decreased=blue, increased=light orange. The estimation of the protein and mRNA concentrations was derived by published datasets and color-coded on a white-green scale for protein concentrations and on a white-yellow scale for mRNA concentrations.

embryos (Lee *et al.*, 2022 [\[1\]](#)); it is required for centriole duplication. Therefore, although centrioles are not assembled, all centriolar proteins are expressed in the full-grown oocyte and these components serve as a stock for the biogenesis of the centrioles of the embryo. Our results suggest that specific phosphorylations of proteins regulating centriole assembly could prevent the *de novo* establishment of centrioles in the oocyte.

PCM proteins

The core element of the PCM is the γ -tubulin ring complex (γ -TuRC) which is the basic element for nucleation of microtubules from microtubule-organizing centers (MTOCs) (Oakley & Oakley, 1989 [\[2\]](#)). It includes γ -tubulin that is combined with other proteins known as GCP2/tubgcp2 to GCP6/tubgcp6. Some PCM proteins or regulators of PCM assembly have been described in functional studies to be present in the oocyte and then recruited to the zygotic centrosome as maternal components. These include γ -tubulin (Stearns & Kirschner, 1994 [\[3\]](#); Gard, 1994 [\[4\]](#); Félix *et al.*, 1994 [\[5\]](#)), pericentrin (Dichtenberg *et al.*, 1998 [\[6\]](#)), Nek2B (Fry *et al.*, 2000 [\[7\]](#); Uto, 2000 [\[8\]](#)), SSX2IP (Bärenz *et al.*, 2013 [\[9\]](#)), TPX2, PRC1, Kif4A, Eg5/Kif11, CLASP1/Xorbit and Kif22/Xkid, CEP152 (Hatch *et al.*, 2010 [\[10\]](#)), NEDD1 (Liu & Wiese, 2008 [\[11\]](#)), Maskin/TACC3 (Stebbins-Boaz *et al.*, 1999 [\[12\]](#); Peset *et al.*, 2005 [\[13\]](#)), XMAP215/CKAP5 (Popov *et al.*, 2002 [\[14\]](#); Kinoshita *et al.*, 2002 [\[15\]](#)) and Nercc1/NEK9 (Roig *et al.*, 2005 [\[16\]](#)).

Most of the PCM proteins were detected in our proteomic dataset and either accumulate or stay constant during meiotic divisions (Table 1 [\[1\]](#)). Our dataset highlights that γ -tubulin accumulates during meiotic divisions at the time the oocytes enter MII (Table 1 [\[1\]](#)-table supplement 1C). GCP2 and 4 are expressed at a lower level than γ -tubulin (Table 1 [\[1\]](#)), in agreement with the expected stoichiometry of GCPs- γ -tubulin in the γ -TuRC (Sulimenko *et al.*, 2022 [\[17\]](#)). The phosphorylation of γ -TuRC components is known to affect the stability and the activation of the complex (Sulimenko *et al.*, 2022 [\[17\]](#)). We detected the phosphorylation of GCP3 at S814 (Class IV) and GCP6 at S1173 (Class III) (Table 1 [\[1\]](#)-table supplement 1D). This second phosphosite is well conserved in the human protein and targeted by Plk4, which regulates centriole duplication (Bahtz *et al.*, 2012 [\[18\]](#)). These results indicate that the oocyte, despite the absence of centrioles, prepares a functional centrosomal machinery by accumulating and/or phosphorylating some of the PCM proteins. Apart of the γ -TuRC complex, we found that 10 other PCM proteins are phosphorylated during meiotic maturation (CEP192, SSX2IP, CEP41, TACC3/Maskin, Nucleophosmin, Katanin, Kif11, Kif15/Eg5, Numa and Dynein). CEP192 is a coiled-coil scaffolding protein that recruits γ -TuRC, Aurora-A and Plk1 to the centrosome for its maturation. Our phosphoproteome reveals that CEP192, a Class II protein, is phosphorylated at 7 sites between prophase and MI, 5 of them corresponding to Cdk1 phosphorylation sites (Table 1 [\[1\]](#)-table supplement 1E). Plk1 is recruited to its many different locations in the cell through its Polo-Box-Domain (PBD), which binds to phosphorylated S-S/T(p) motifs (Elia *et al.*, 2003a [\[19\]](#), 2003b [\[20\]](#); Lee *et al.*, 1998 [\[21\]](#); Reynolds & Ohkura, 2003 [\[22\]](#)). Two of the CEP192 sites identified in our phosphoproteome, S991 and S1227, correspond to such motifs. Therefore, by attracting Plk1, CEP192 phosphorylation could prepare the reconstitution of a functional centrosome at fertilization.

Assembly of microtubule spindles and microtubule dynamics

Xenopus prophase oocytes are inefficient in their ability to support microtubule polymerization (Jesus *et al.*, 1984 [\[23\]](#); Gard & Kirschner, 1987 [\[24\]](#); Heidemann & Kirschner, 1975 [\[25\]](#)), although microtubules are formed at the cortical and nuclear envelope level, where γ -tubulin foci are also localized (Gard, 1994 [\[4\]](#); Jesus *et al.*, 1988 [\[26\]](#)). The ability to assemble microtubule asters is acquired at the time of NEBD, when a giant disk-shaped MTOC generates a large microtubular monoaster at the basal region of the disintegrating nucleus (Huchon *et al.*, 1981 [\[27\]](#); Gard, 1994 [\[4\]](#)). Therefore, MTOC and microtubule-associated proteins (MAPs) must be finely regulated by post-translational modifications at time of NEBD. The transient microtubule monoaster transports the meiotic chromosomes to the animal cortex, where the first meiotic spindle is assembled (Huchon *et al.*, 1981 [\[27\]](#); Gard, 1992 [\[28\]](#)). Due to the absence of centrioles, meiotic spindles are barrel-shaped

and are formed by centrosome-independent microtubule nucleation originating from the chromosomes and involving chromatin and the Ran GTPase (Gruss & Vernos, 2004 [↗](#)), much like spindles of vascular plant cells. Moreover, the two successive spindles are differentially regulated. The MI spindle is a transient and dynamic structure while the MII spindle is stable for hours, anchored to the plasma membrane until fertilization (Gard, 1992 [↗](#)). Therefore, it is possible that MAPs are specifically and differentially regulated at MI and MII (Classes III and V). Finally, a major change in microtubule assembly occurs during the transition from meiosis to embryonic mitosis. The assembly of the first embryonic spindle integrates the meiotic chromosome-dependent pathway with the newly formed centrosomes (Cavazza *et al.*, 2016 [↗](#)). Thereafter, subsequent embryonic spindles depend solely on the microtubule nucleating activity of the centrosomes (Kirschner, 1986 [↗](#)). All these critical transitions in the way of assembling a microtubular spindle depend on the proteins stored in the oocyte.

At least 5 microtubule regulatory proteins undergo specific patterns of phosphorylation that deserve attention (Figure 8 [↗](#)). Katanin (KATNB1), which is responsible for the majority of M-phase severing activity in *Xenopus* eggs and is activated by Cdk1 (McNally & Thomas, 1998 [↗](#)), displays a bi-modal pattern of phosphorylation, increasing at the time of formation of the monoaster-MTOC and MI spindle, decreasing during MI-MII transition and increasing again at MII (Figure 8 [↗](#)). Kif11/Eg5 phosphorylation occurs at T936 and T937 starting NEBD (Figure 8 [↗](#)). Both phosphosites were reported in a broad range of models, and the second site was shown to be phosphorylated by Cdk1 (Blangy *et al.*, 1995; Cahu *et al.*, 2008 [↗](#); Giet *et al.*, 1999 [↗](#)) (Figure 8 [↗](#)). Kif11/Eg5 controls the relative stability of bipolar versus monopolar organization of spindles in *Xenopus* egg extracts (Mitchison *et al.*, 2005 [↗](#)) but also centrosome disjunction and/or separation. Hence, Kif11 is a good candidate to be an important player regulating the original changes in microtubule dynamics and organization of the two successive spindles. Kif15, also known as XKlp2, a plus end-directed kinesin protein required for centrosome separation and maintenance of spindle bipolarity in *Xenopus* egg extracts (Boleti *et al.*, 1996 [↗](#)), is phosphorylated at S681 and S692 during meiotic maturation. S681 is rather phosphorylated at MI while S692 is specific of MII (Figure 8 [↗](#)), suggesting a potential distinct regulation of Kif15 on microtubule dynamics at MI and MII. Dyneins are a family of motor proteins that move to microtubules in an opposed direction to kinesins. Among this family, the two paralog genes dynein light intermediate chain (DynC1LI1 and DynC1LI2) play important roles in mitosis, including positioning the spindle and focusing the MTs into poles. Our phosphoproteome reveals multiple phosphorylations of these proteins, starting in MI and remaining constant or progressively increasing until MII, as for S197 of DynC1LI1 (S207 in human) (Figure 8 [↗](#)). This agrees with previous data showing that S197 is phosphorylated by Cdk1 (Dell *et al.*, 2000 [↗](#); Addinall *et al.*, 2001 [↗](#)) and that DynC1LI becomes hyperphosphorylated at the time of NEBD and remains hyperphosphorylated throughout the rest of meiosis (Huang *et al.*, 1999 [↗](#)). Additionally, our phosphoproteome reveals that 3 sites (495, 498 and 501) are phosphorylated around NEBD and get dephosphorylated after MI (Class III), hence being correlated with the 1st meiotic division and not the 2nd one, while 6 other sites (197, 396, 409, 446, 452, and 467) are phosphorylated both in MI and MII (Class II) (Figure 8 [↗](#)). Dynein is required for organization of the original microtubule/MTOC array that is organized at NEBD, together with XMAP215, XKCM1 and Numa (Becker *et al.*, 2003 [↗](#)). The formation of this oocyte original monoaster/MTOC could involve the dynein phosphorylations that transiently take place precisely at this period.

Strikingly, MI and MII spindles segregate chromosomes and chromatids in a very different way. In MI, the homologous chromosomes are segregated, whereas in MII the sister chromatids are displaced in the daughter cells, due to a specific organization of kinetochores and regulation of cohesins. In MI, arm cohesins are phosphorylated and degraded whereas centromeric cohesins are protected from cleavage by Sugoshin (Sgo1)-PP2A. This allows the separation of chromosomes, but chromatids are still attached together. In MII, centromeric cohesins are fully degraded, allowing the segregation of sister chromatids, as during mitosis (Wassmann, 2022 [↗](#)). How a fraction of cohesins is protected from cleavage in MI but not in MII is still not entirely clear since

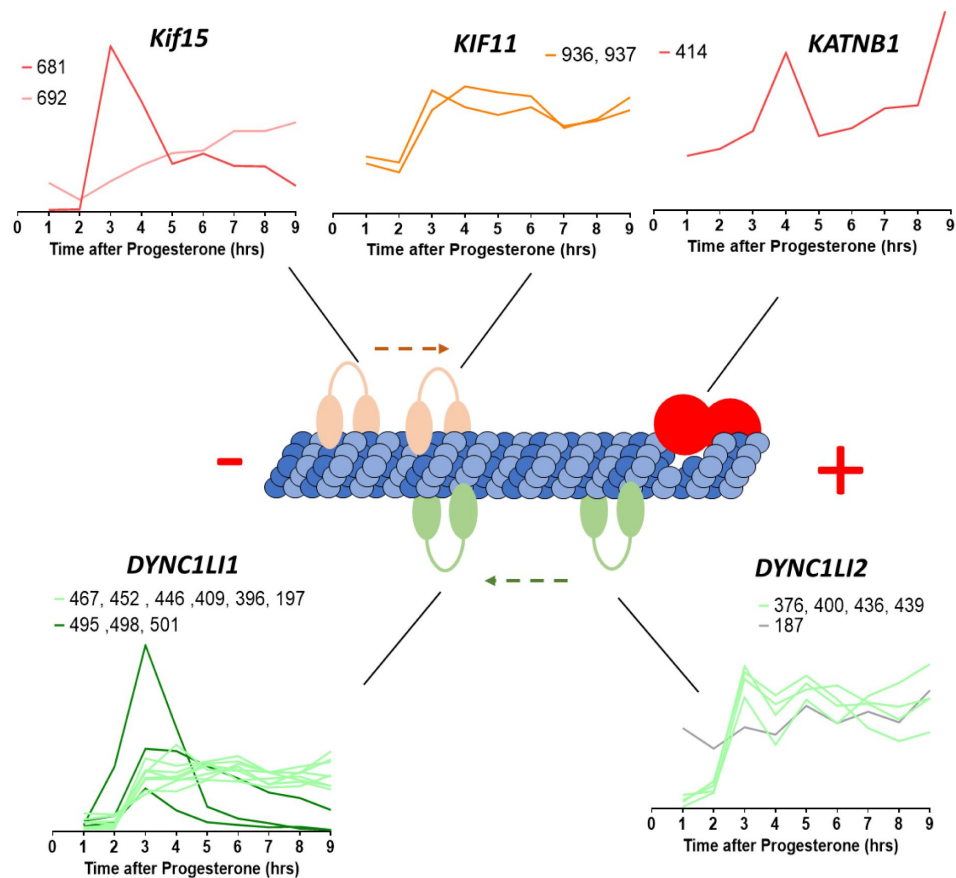


Figure 8.

Microtubule-interacting proteins phosphorylation during meiotic divisions. Graphic representation of the phosphorylation dynamics of kinesins (Kif15, Kif11), dyneins (DYNC1LI1, DYNC1LI2) and katanin (KATNB1) during meiotic divisions. The sites of the phosphorylation detected are marked above each panel.

the phosphorylation of cohesins and their partners, which are not very abundant proteins, is difficult to be detected with conventional shotgun phosphoproteomics. Notably, our phosphoproteomic studies highlight new sites differentially phosphorylated between MI and MII within these proteins. Sgo1 and two components of the cohesin complex, Pds5b and Wapl, are increasingly phosphorylated during meiotic divisions (Class IV) (**Figure 8** [figure supplement 1](#)). Noteworthy, S1069 of Wapl is a putative Plk1 site (Grosstessner-Hain *et al*, 2011 [figure supplement 1](#); Kettenbach *et al*, 2011 [figure supplement 1](#)). By highlighting new sites differentially phosphorylated between MI and MII, our phosphoproteomic studies provide clues to elucidate the still unknown mechanisms of the meiotic mechanism of cohesion protection-deprotection.

Discussion

The amphibian oocyte and egg have long been a source of inspiration and a source of experimental opportunity for important areas of biology, such as nuclear organization, spindle formation, cell polarity, fertilization, mitosis and meiosis, endocrine signaling and intracellular signal transduction. As experimental embryology gave way to genetic approaches in the late 20th century to study embryonic development, the attractiveness of amphibian oocytes declined.

The original advantages of the amphibian oocyte are now becoming more obvious, as it becomes important to link complex cellular processes, like cell division, to signal transduction. These processes are difficult to study because they occur at a single cell or even at subcellular level. Indeed, many important cell-division related processes involve transient changes occurring in specific compartments of the cell, like the endoplasmic reticulum, the mitotic apparatus, the nuclear and plasma membranes, or the cytoskeleton. Additionally, meiotic and mitotic divisions occupy a very short fraction of the life span of germ and somatic cells, making it hard to study them in cultured cells. Frog oocytes and eggs show exquisite synchrony of cell division, allowing to have large cohort of cells at the same stage of the cell cycle. Therefore, we and others have returned to the awesome power of the amphibian oocyte and egg to study the signal transduction, intracellular posttranslational changes surrounding the onset of meiotic and mitotic divisions and fertilization at a single cell level.

Importantly, the ingenious and sensitive tools developed to study the transcriptional regulation are of little use in the study of a cell type that does not transcribe RNA and of events that are regulated post-translationally, as meiotic and early embryonic divisions. For a long time, we lacked general tools for identification and measurement of the activity of the regulators of these processes, impairing our ability to connect the morphological changes previously described by histology to the respective molecular events. One manner to investigate these pathways is using pharmacological inhibitors, but their non-specificity and the complexity of pathways pose serious limitations. For years, phosphorylation events have been extensively studied using candidate-based approaches, as site-specific mutagenesis and phosphospecific antibodies. These targeted-approaches have several caveats. The selection of the putative sites of phosphorylation is based on *in silico* evidence, limiting the analysis to proteins bearing canonical consensus sequences and hence our knowledge of novel phosphorylation motifs. When multiple putative phosphorylation sites are present in the same protein, mutagenesis of individual sites or combinations of sites becomes too complex, impairing identification of the phosphorylation sites used *in vivo* and their interdependency or redundancy. Moreover, the time-consuming production of phospho-specific antibodies is not scalable to high throughput applications. Mass spectrometry has the potential of shedding light on these intricate networks. However, the current tools are unable to accurately quantify the level of proteins and phosphorylated proteins at the single cell level -except for the frog oocytes and eggs, thanks to their exceptional size and high protein content.

Oocyte meiotic maturation is a key biological process. First, it transforms the oocyte into a fertilizable cell at the origin of life of any animal. Second, since meiosis derives from mitosis, thanks to cellular adaptations that were necessary to produce a haploid gamete (preventing an S-phase between the two divisions, modifying the functioning of the spindle during meiosis I, establishing division arrests needed for the oocyte to grow or to wait for sperm, performing asymmetrical divisions to save nutrient reserves, etc.), its study provides the opportunity to understand how cell division can adapt and diversify in living systems, depending on specific cellular organization and specialized functions.

In this paper, we have applied the current state of the art of quantitative mass spectrometry to oocyte maturation and meiotic divisions using quantitative proteomics and quantitative phosphoproteomics. We have temporally correlated events with oocyte maturation induced naturally by progesterone to *Xenopus* full-grown oocytes and watched the progression through meiosis I to the natural arrest at the second meiotic metaphase. We have been able to correlate the proteomic measurements with the physiological changes as the oocyte, as it goes from an early prophase I state through to a second meiotic metaphase state. We find dramatic changes in the oocyte nucleus and the meiotic spindles and drastic modifications to the standard cell cycle. These dramatic changes are choreographed principally by signaling pathways employing kinases and phosphatases. In all animals, meiotic maturation relies on the dynamics of phosphorylation events, either during the signaling pathway that leads to Cdk1 activation or during the Cdk1 downstream period that orchestrates the structural events of the two meiotic divisions. The candidate-based approaches have provided a global scheme of this process based on the knowledge of a few kinases that have been extensively studied for years. However, this scheme still does not provide a very comprehensive picture of the meiotic process, since it does not take into account hundreds of unknown critical kinases, substrates and phosphosites. Our quantitative proteomics and phosphoproteomics approach allowed to obtain a much deeper and comprehensive dynamic picture of the phosphorylated sites during meiotic maturation. This has enabled us to describe and support with biochemical precision the various processes in the oocyte, determine the relative timing of various site phosphorylations and molecular processes, thereby challenging or suggesting the epistatic sequences of specific molecular pathways. Furthermore, we confirmed the importance of certain previously known phosphosites, identified the likely kinases or in some cases the kinase families responsible for substrate phosphorylation, and established connections between phosphorylation events and intracellular remodeling, as well as regulatory mechanisms such as translation and protein degradation.

In addition to explaining the order of diverse events of chromatin remodeling, nuclear organization, plasma membrane transformations, cytoskeletal changes of meiosis, our proteomic and phospho-proteomic datasets should provide a large number of experimental avenues for exploring similar events in both mitotic and meiotic stages of diverse organisms. This molecular data will enable us to better understand regulatory pathways involving chromosome behavior, spindle formation, nuclear and plasma membrane regulation, and signal transduction and derive insights far beyond the scope of meiosis and oocyte maturation. One such insight might come from a closer analysis of the quality control mechanisms in the oocyte. Both nucleopore remodeling (King *et al.* 2019 [DOI](#)) and E2/E3 ubiquitin ligases are involved in housekeeping and quality control, specifically in the removal of aggregated and orphan complex proteins, as the germ cell of an organism prepares to become the next generation

Methods

Oocytes collection and *in vitro* maturation

The research with *Xenopus laevis* was performed under the oversight of the Harvard Medical Area Institutional Animal Care and Use Committee. Mature *Xenopus laevis* females were used after priming with pregnant mare serum gonadotropin (PMSG, 50 IU per frog). A sample of ovary tissue was surgically removed, rinsed in 1× Marc's modified Ringer's (MMR) and placed in OR2 buffer (83 mM NaCl, 2.5 mM KCl, 1 mM CaCl₂, 1 mM MgCl₂, 1 mM Na₂HPO₄, 5 mM HEPES, pH 7.8) and oocytes were manually defolliculated. Stage VI oocytes were selected and kept in dishes with agar beds to avoid sticking to the dish. Maturation was induced by incubation at 18 °C with 10 µg/mL progesterone in OR2 buffer.

Eggs were flash frozen with liquid nitrogen in bunches of ten or fifty for protein and phosphoproteomics measurements respectively. All samples were taken in triplicates, each biological replicate was performed with different frogs. The first sample was taken from untreated oocytes at the time of progesterone treatment, the second timepoint 2 hours after progesterone treatment, with hourly timepoints past that, up to 9 hours post treatment.

Embryo lysis and digestion into peptides

For each condition the embryos were lysed at a ratio of 6 µL buffer/embryo in the same buffer as (Presler *et al*, 2017 [\[4\]](#)) (0.25 M sucrose, 1% replacement NP-40, 10 mM EDTA, 25 mM HEPES, 1 mM TCEP, 10 µM Combretastatin, 10 µM Cytochalasin D, Roche Protease Inhibitor Mini EDTA free, and Phosstop inhibitor (Roche) at pH 7.2). The yolk was removed by spinning the samples at 4,000 g for 4 minutes at 4°C. Both the supernatant and all lipids were transferred for a new tube. The concentration of HEPES was raised to

0.1 M and SDS was added to 20%. After addition of DTT to 5 mM, the samples were denatured in a 60°C water bath for 20 minutes. After the samples temperatures return to room temperature, N-Ethylmaleimide (NEM) was added to 15 mM, tubes were mixed thoroughly, and the reaction proceeded for 1 hour. Then additional DTT was added to 5 mM to quench the excess NEM. The samples were flash frozen and stored at -80°C until the methanol chloroform precipitation (Wessel & Flügge, 1984 [\[5\]](#)). Protein was precipitated by methanol-chloroform precipitation. Due to the large volume of lysate (~ 1 mL), the precipitation was performed in glass 15 mL Corex tubes (No. 8441). The tubes were cleaned for a two- hour chromic acid incubation and rinsed extensively in filtered water. Then they were autoclaved and before use for the precipitation, they were rinsed with HPLC H₂O. The centrifugations were done in a JA 50 30 Ti rotor in a Beckman Avanti J-30I centrifuge at 21,000 g for either 6 minutes or 15 minutes. After the last supernatant was removed, the pellets were air dried for 15 minutes and resuspended in 0.6 Ml of 6 M guanidine, 50 M EPPS pH 8.5 and transferred to a 2 mL eppendorf tube. The samples were incubated for 5 minutes at 65°C and the protein amount was determined by a BCA Assay. The solution was diluted to 2 M guanidine with 10 mM EPPS pH 8.5. Lysyl endopeptidase (FUJIFILM WAKO (129-02541) resuspended with HPLC H₂O to ~2 mg/mL) was added to a concentration of 20 ng/mL and the initial digestion proceeded for 12 hours at room temperature with gentle shaking. The samples were further diluted to 0.5 M guanidine with 10 mM EPPS pH 8.5 and Trypsin protease (Sequencing Grade Modified Trypsin, Promega) was added to a final concentration of 10 ng/mL along with an additional 20 ng/mL of LysC. The samples were incubated in a 37°C room for 8 hours on a nutator.

Phospho-peptide enrichment

To carry out the phosphoproteomic analysis, we made the conventional steps of digesting the proteins from each time sample with trypsin and specifically labeling them with isobaric tags (TMT-10 reagents). Tagged samples and the phosphopeptides were enriched on an IMAC column. Importantly, we chose to multiplex peptides before the phosphopeptide enrichment to improve data quality. There is a tradeoff of decreased yield and therefore depth, as it is not economical to

label more than a few milligrams of material. We used 2.5-4 mgs of TMT-labeled peptides per replicate, enriched with 5 μm Titanium Dioxide microspheres (GL Sciences 5020-75000) and fractioned as previously described (Presler *et al*, 2017 [↗](#)). A typical yield of 50-80 μg of peptides eluted from the column, with a median phosphopeptide enrichment of $\sim 80\%$.

Phosphatase treatment of *Xenopus* samples prepared for parallel phospho-enrichment

We adapted the method that we used to phosphatase-treat digested peptides (Presler *et al*, 2017 [↗](#)) for the larger samples used here for the parallel phospho-enrichment. We chose to phosphatase treat when proteases were present in the samples (but after the protease digestion sequence described above) to enable the de-phosphorylated peptides to have the cleavage pattern expected if they were present initially in the de-phosphorylated state. We used the heat-labile Alkaline Shrimp Phosphatase (Affymetrix Inc, USA). We used a 3k Amicon Centrifugal Filter to exchange the enzyme storage buffer to 10 mM EPPS pH 8.5 instead of Tris-HCl pH 7.5. We also concentrated the phosphatase to between 3-4 U/ μL (confirmed by p-nitrophenyl phosphate assay). We added an estimated 833 units of phosphatase to two of the samples built to contain representative mixture of peptides (TMT 131N, channel 10: 100 oocytes collected at the latest timepoint, and TMT 131C, channel 11: 50 oocytes collected at the first timepoint combined with 50 oocytes collected at the last timepoint). To samples that were not phosphatase-treated, we added the same volume of blank phosphatase buffer: 5 mM EPPS pH 8, 50% glycerol. To all samples, we added MgCl_2 to 10 mM and EDTA to 0.1 mM (to chelate metals that might compete with the magnesium necessary for enzymatic activity). All the samples were incubated with gentle shaking at room temperature for 12 hours. The phosphatase was then inactivated by incubation with gentle shaking in an air-incubator at 65°C for 15 minutes. Peptide fractionation and TMT-MS3 LC/MS were carried out as done previously (Presler *et al*, 2017 [↗](#)).

Computing phospho-occupancy systematically

We grouped phospho-peptides by unique phospho-site identifiers. In order to rigorously reflect the measurement accuracy and to take advantage of many measurements of the same phospho-site (including on peptides with different missed-cleavages or oxidized methionines), we used BACIQ (Peshkin *et al*, 2019 [↗](#)) to compute the estimated trend with confidence intervals. Then, we created phospho-site sets to connect phospho-sites on residues that could be on the same peptide. We made these assignments based on whether the individual phospho-sites were ever measured with another phospho-site as a composite phospho-site. In order to determine occupancy, we needed to measure the phospho-site as a different form (usually not-phosphorylated). To do this, we identified non-phosphorylated peptides measured from the same experimental conditions that include the residue(s) of the phospho-site. Again, multiple non-phosphorylated peptides can include the residues of interest and the same peptide sequences can be measured multiple times. Therefore, we again use BACIQ (Peshkin *et al*, 2019 [↗](#)) to aggregate these measurements in a trend with confidence intervals. The MATLAB implementation is freely available at https://github.com/elizabeth-van-itallie/phospho_occupancy_matlab [↗](#).

Mass spectrometry data mapping and analysis

Peptide-Spectra matches were performed as previously described (Sonnett *et al*, 2018 [↗](#)). Assignment of MS2 spectra was performed using the SEQUEST (Eng *et al*, 1994 [↗](#)) algorithm by searching the data against the appropriate proteome reference set acquired from Xenbase (*Xenopus laevis* assembly 9.1) along with common contaminants: human keratins and trypsin. The target-decoy strategy was used to construct a second database of reversed sequences that were used to estimate the false discovery rate on the peptide level. SEQUEST searches were performed using a 20 ppm precursor ion tolerance with the requirement that both N- and C-terminal peptide ends are consistent with the protease specificities of LysC and Trypsin. TMT (+229.162932 Da) was set as a static modification on N-termini and lysines peptides, and N-ethyl maleimide (+125.047679

Da) was set as a static modification on cysteine residues. Oxidation of methionine (+15.99492 Da) was set as a variable modification. A peptide level MS2 spectral assignment false discovery rate of 1% was obtained by applying the target-decoy strategy with linear discriminant analysis. Peptides of 7 amino acids length or longer were ranked by linear discriminant analysis score and were filtered to a ratio of 1% reverses/forwards + reverses. Peptides were assigned to proteins and a second filtering step to obtain a 1% FDR on the protein level was applied. Peptides that matched multiple proteins were assigned to the proteins with the most unique peptides. We only used Isolation Specificity > 0.75 spectra.

Estimation of the absolute abundance of proteins

The absolute protein concentration was estimated according previously published methods (Wühr *et al*, 2014 [↗](#); Peshkin *et al*, 2015 [↗](#)), based here on ion current prorated to the isobarically labeled fractions. The respective sample fractions were estimated using our previously published Bayesian approach (Peshkin *et al*, 2019 [↗](#)), which integrates peptide signal and peptide-level measurement agreement into a maximum likelihood estimate of the true protein ratio and the associated confidence interval (Peshkin *et al*, 2019 [↗](#)). See Appendix 1.

Data availability

The mass spectrometry proteomics raw data and reference set of sequences have been deposited to the ProteomeXchange consortium via the PRIDE partner repository with the dataset identifier MassIVE MSV000094498. In addition, we developed a multi-functional Web portal to release the data and enable interactive interrogation (https://xenopus.hms.harvard.edu/index_oocyte.html [↗](#)). All of the aspects from overall protein trends to individual clusters to specific phosphosites and stoichiometry information are easily available. The portal provides per-protein link to external information about each human protein by symbol from GeneCards (<https://www.genecards.org/> [↗](#)) and respective *Xenopus* protein at Xenbase (<https://www.xenbase.org/xenbase/> [↗](#)).

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Author contributions

Leonid Peshkin: Conceptualization; Methodology; Experiments; Software; Formal Analysis; Data Curation; Writing; Supervision. **Enrico M Daldello**: Formal Analysis; Writing. **Elizabeth Van Itallie**: Software; Formal Analysis; Data Curation; Writing. **Matthew Sonnett**: Methodology; Formal Analysis. **Johannes Kreuzer**: Formal Analysis; Data Curation. **Wilhelm Haas**: Methodology; Formal Analysis. **Marc W Kirschner**: Conceptualization; Methodology; Experiments; Writing; Supervision. **Catherine Jessus**: Writing; Formal Analysis; Supervision. **Marc W Kirschner** and **Catherine Jessus** contributed equally to this work.

Disclosure and competing interests statement

The authors declare no competing interests.

Appendix 1

Estimating the absolute abundance of proteins

Following previously published methods (Wühr *et al.*, 2014 [DOI](#); Peshkin *et al.*, 2015 [DOI](#)), the absolute protein concentration was estimated here based on ion current (denoted I) prorated to the isobarically labeled fractions. The respective sample fractions were estimated using our published Bayesian approach (Peshkin *et al.*, 2019 [DOI](#)), which integrates peptide signal and peptide-level measurement agreement into a maximum likelihood estimate of the true protein ratio and the associated confidence interval (BACIQ). Specifically, we want to construct the following relation applicable to all measured proteins:

$$\log(C_S^P) = A_p \log(Q_S^P) + B_p, \quad Q_S^P = \frac{1}{N_p^T} \cdot F_S^P \cdot I$$

where:

Q_S^P is the isobarically labeled fractions of the total I ion current;

C_S^P is the absolute abundance (in Moles) of the protein p at a stage s ;

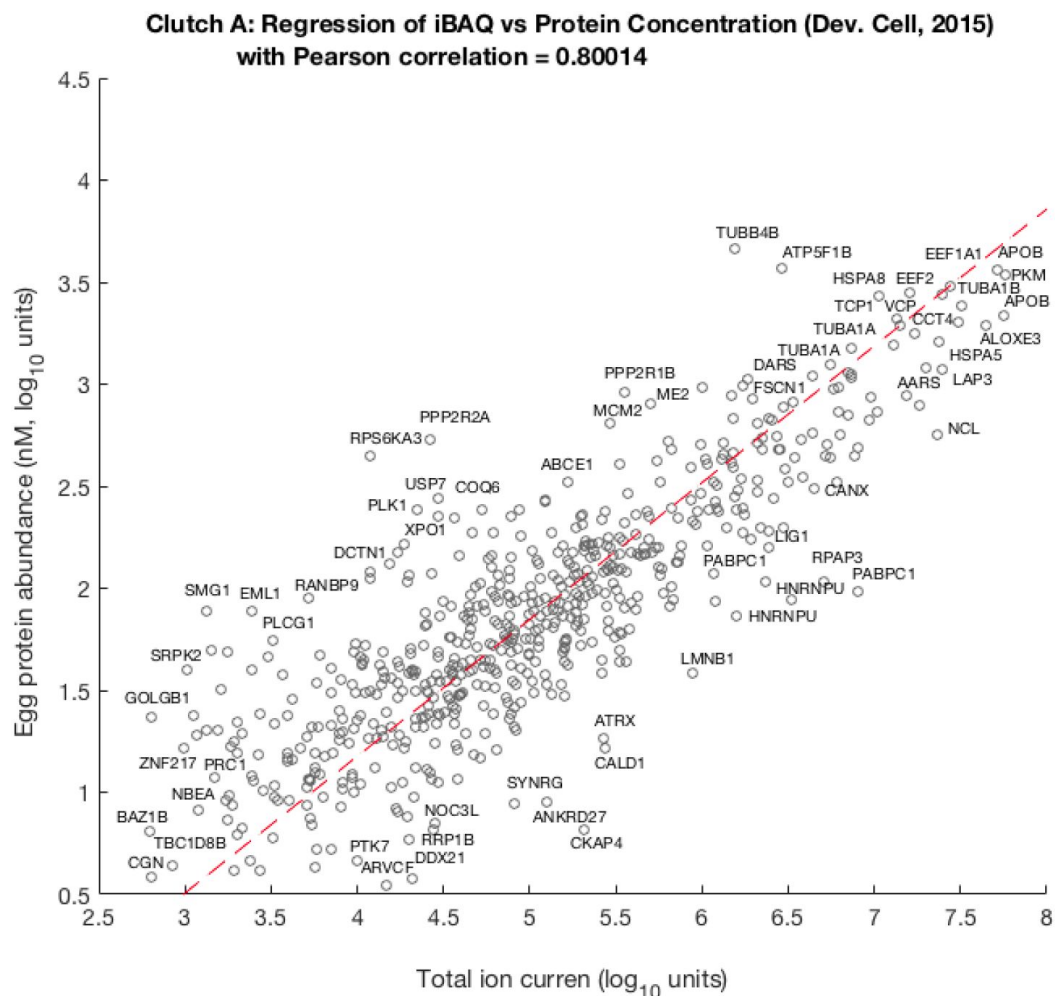
N_p^T is the theoretical number of (unique) peptides obtained by "in-silico digestion" script (available in the supplementary materials) using the following parameters: miss_cleav=2, min_len=7, max_len=57, enz='[KR]' and FASTA file of the considered protein sequences;

F_S^P is the fraction of the total signal observed in the isobaric peptide signal quantitation channels corresponding to the sample from stage S ;

I is the total ion current reported using only unique peptides as obtained from *cq_area_sum* column of export from the MS platform;

A_p and B_p are the linear regression parameters.

Using the calibration protein data, we can fit (using two-way regression to avoid any bias towards one of the variables) a linear function (1) in log/log space and extrapolate from it by estimating the regression parameters A_p and B_p universal for all proteins. To calibrate the protein measurements, we used previously published (Peshkin *et al.*, 2015 [DOI](#)) concentrations C_S^P for 5960 proteins in *Xenopus laevis* egg. The parameters A_p and B_p are obtained using symmetrical linear regression analysis (in log-log space; **Fig. 1** [DOI](#)) between C_{egg}^P (concentration in the egg stage post-fertilization) and Q_S^P (isobarically labeled fractions of the total I ion current). Further, we only used in the regression analysis proteins with highly confident mapping between proteins in both data sets by sequence homology. The highly confident mapping - reciprocal best hit (RBH) by sequence homology is defined by HMMER3 pipeline (Savova *et al.*, 2017 [DOI](#)) with the perfect E-values both ways) matches of proteins between data published in (Peshkin *et al.*, 2015 [DOI](#)) and proteins in the current study. This filter leaves 617 proteins used in regression analysis as shown in **Fig. 1** [DOI](#) for one of three biological repeats (Clutch A).

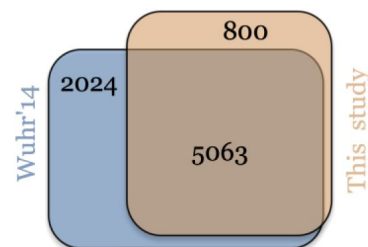


Methods Figure 1.

Scatter plot of per-channel pro-rated ion current Q^P_S (isobarically labeled fractions of the total I ion current) adjusted by theoretical number of tryptic peptides against egg protein abundance as previously published by (Peshkin et al, 2015). Pearson correlation between these quantities is ca. 0.8 the best linear fit leads to the following relation:

$$\log(C^P_S) = 0.67 * \log(Q^P_S) - 1.5$$

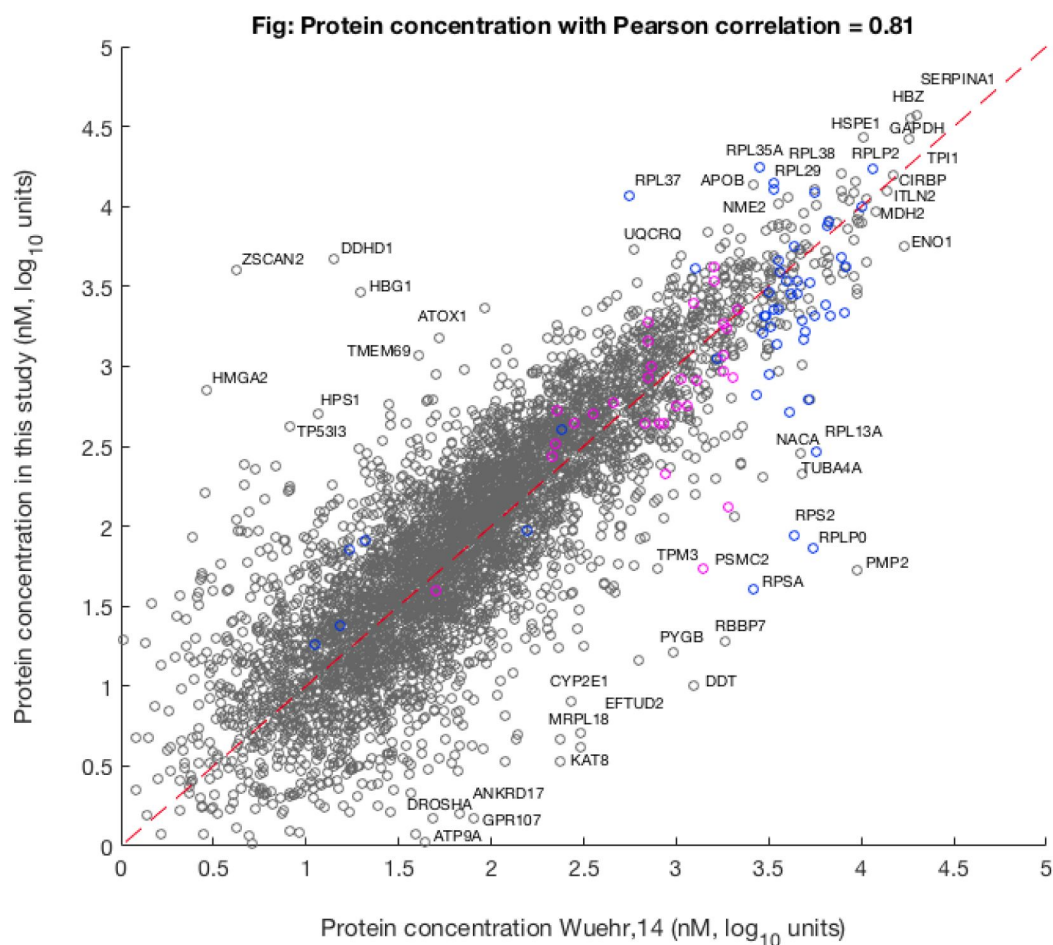
Using the fitting parameters, we converted mass spectrometry measurements into absolute abundance for all proteins detected in the experiments for three biological repeats (Clutch A, B and C). To additionally validate the identification and estimated concentrations of proteins in this study, we performed the comparison with previously published data by (Wühr et al, 2014) matching proteins via respective human gene symbol as assigned via reciprocal HMMER3 run. **Fig. 2** shows Venn diagram illustrating that vast majority of proteins we identified (5063 or 86%) were also detected and confirmed in previously published data, while there are 800 novel proteins. Our current study was done for quantitative comparison across multiple timepoints and thus, being a labeled mass spectrometry experiment, could not be as deep as the unlabeled single sample unlabeled mass spectrometry study by (Wühr et al, 2014).



Methods Figure 2.

Venn diagram comparing proteins identified in this study to previously published by (Wühr *et al*, 2014). This study has 800 unique proteins versus 2024 unique proteins in the data by (Wühr *et al*, 2014). There are 5063 proteins in common.

The 5063 proteins common between our data and (Wühr *et al*, 2014) are used to make comparison of proteins concentrations as shown in Fig. 3. Pearson's correlation between previously published protein concentrations and obtained oocyte proteins in this study is 0.81. The maximum deviation of the extrapolated value from previously estimated value is ca. 3 nM. The differences between studies might result from the biological variation between frogs, difference in gene symbol assignment between studies or differences in the sample preparation.

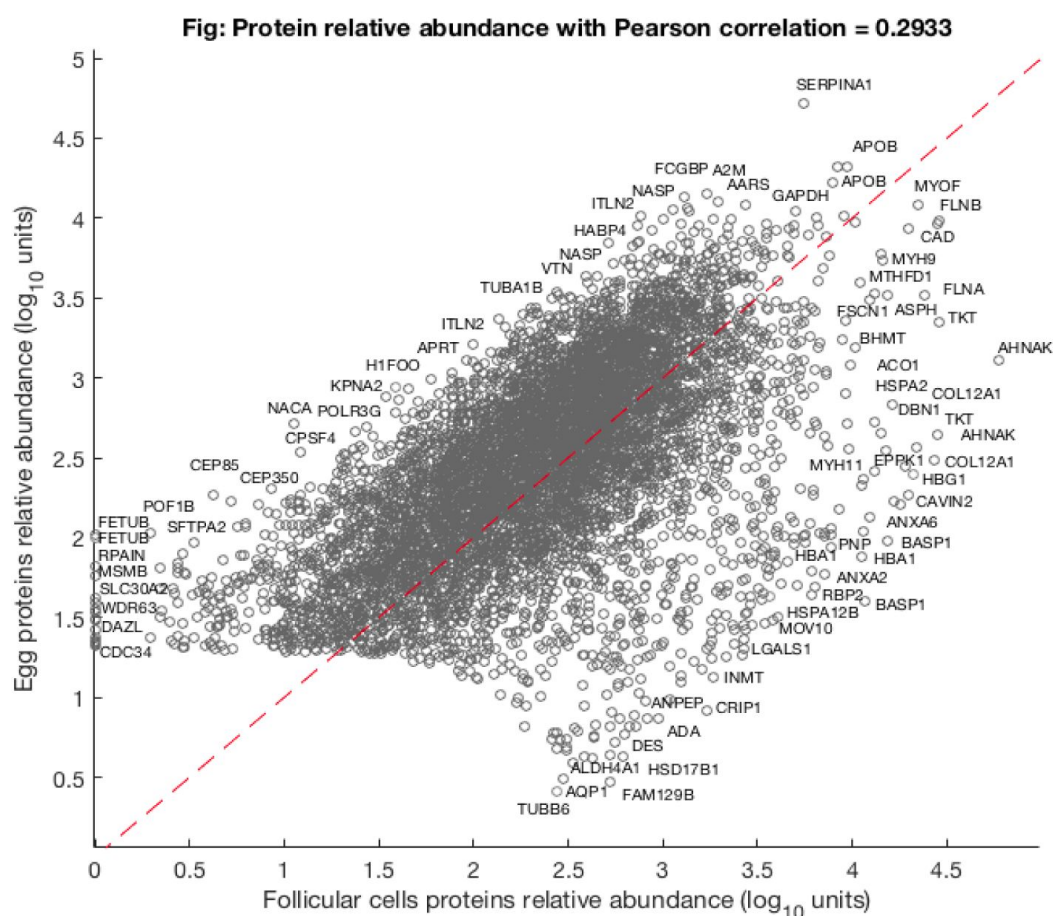


Methods Figure 3.

Comparison of average concentration (i.e., mean value of 3 different oocyte samples) with egg data published previously by (Wühr et al, 2014). The magenta points show the location of proteasome units and blue points show the ribosomal proteins.

Identification and exclusion of follicular cells proteins

Oocytes are surrounded in ovary by an envelope of follicular cells, blood vessels and connective tissue. To study the dynamics of protein expression in oocytes after progesterone stimulation, we physically removed the follicular matter with forceps. This process is imperfect, leaving some amount of follicular matter attached to the oocytes. To control for admixture of the follicular layers, the removed follicular material was separately collected and profiled as one of the samples in mass spectrometry. The average relative protein abundance in oocyte was compared to follicular protein expression as shown in Fig. 4 across a total of 7062 proteins.

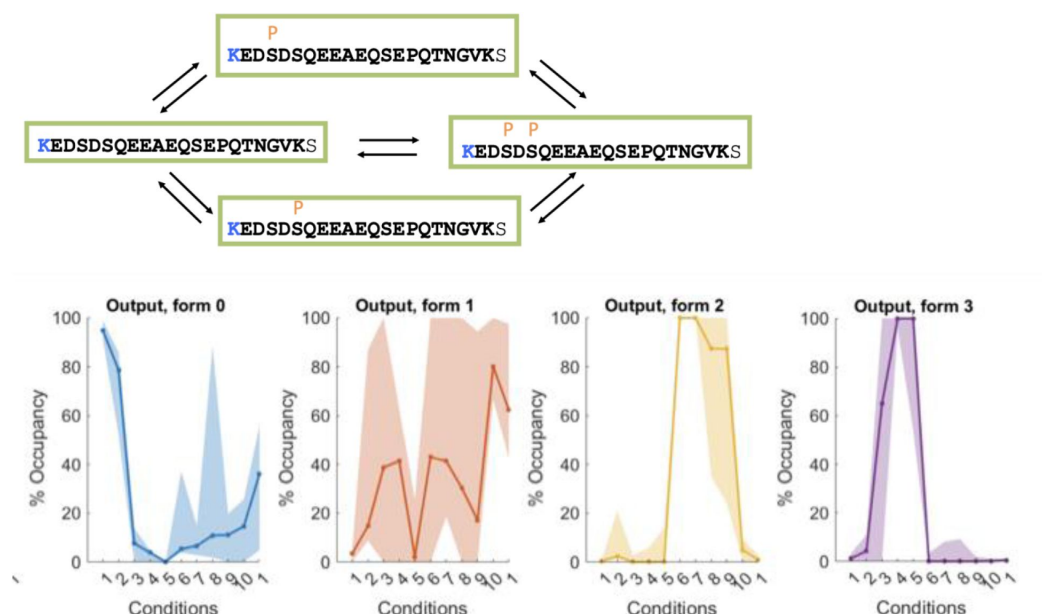


Methods Figure 4.

A scatter plot of a relative protein abundance in oocyte against abundance in follicular tissue. The low abundant proteins are naturally masked in the follicular sample as they are contrasted to 10 oocyte samples, thus severely underrepresented in a pooled sample.

Fig. 4 shows oocyte specific genes FETUB and H1FOO as well as the follicular cells specific proteins HBG1 and TUBB6. The cut off for selecting “exclusively follicular proteins” was somewhat arbitrarily selected at 0.6 orders difference in log-log space or roughly 4-fold enrichment in the follicle taking into account that oocyte material is impure and itself contains substantial follicular residuals. These over- expressed 648 follicular proteins were removed from the oocyte protein analysis since the dynamics of these proteins reflects the artifactual fluctuation in presence of the remaining follicular cells in oocyte samples.

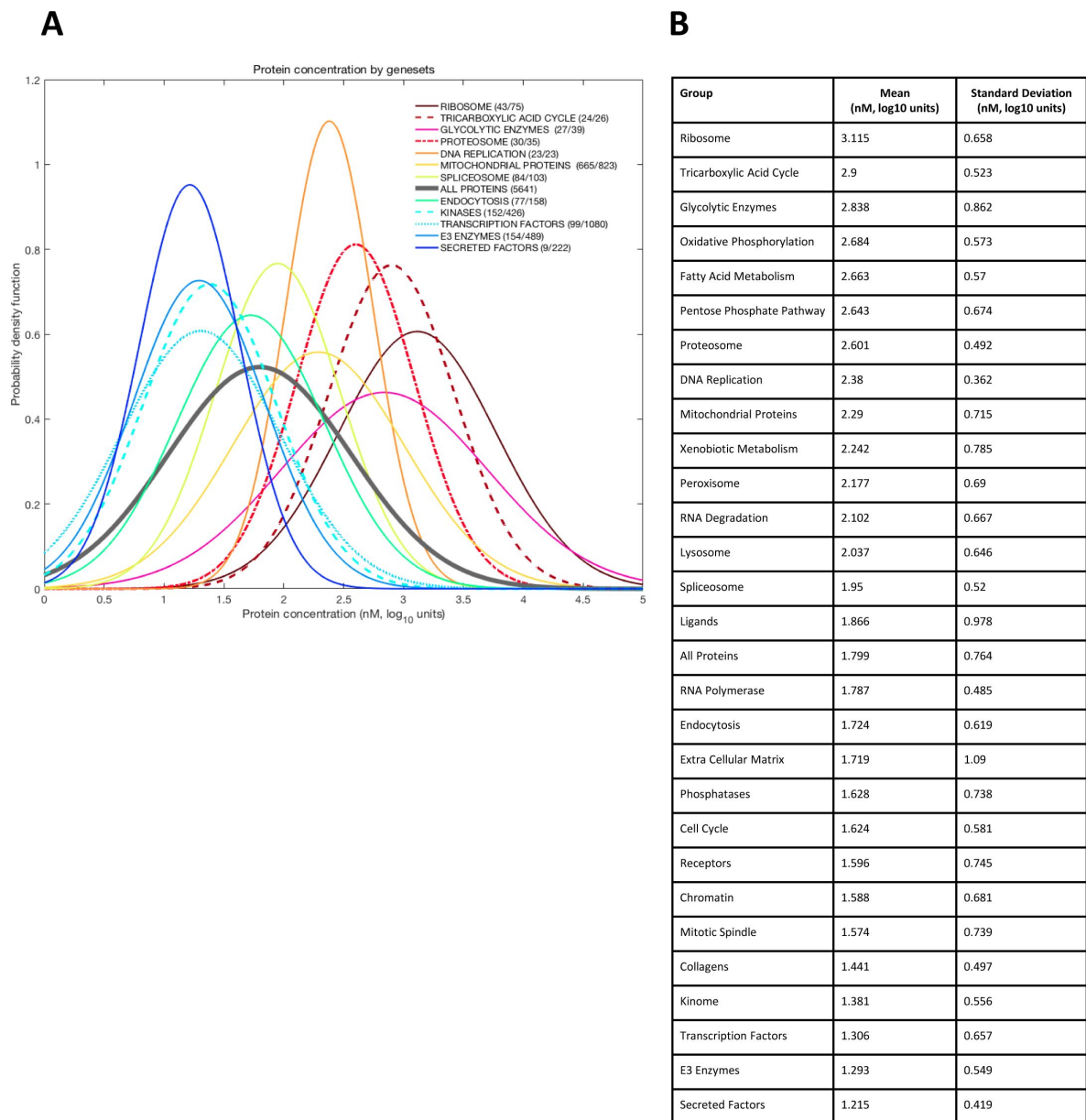
Occupancy Calculation



Methods Figure 5.

An illustration of occupancy calculation for a single peptide where four phospho- forms are measured: the de-phosphorylated, singly phosphorylated on a serine in one of two alternative positions, and doubly phosphorylated on both serines. All interconversions (A) across four forms are allowed by our calculation. The resulting stoichiometry and respective confidence intervals (B) estimated for each form following (Presler et al, 2017) and BACIQ (Peshkin et al, 2019). Points 1- 9 are the timepoints post progesterone stimulation, the two rightmost points are protease-treated samples.

Appendix 2

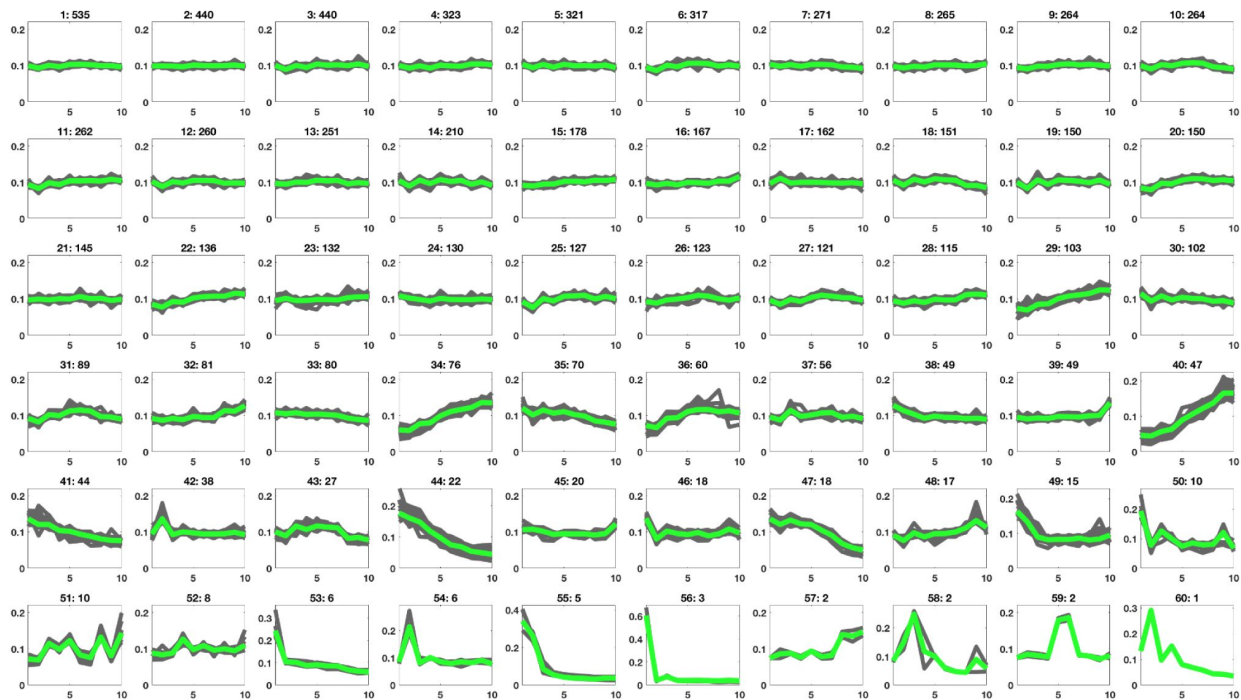


Appendix 2.

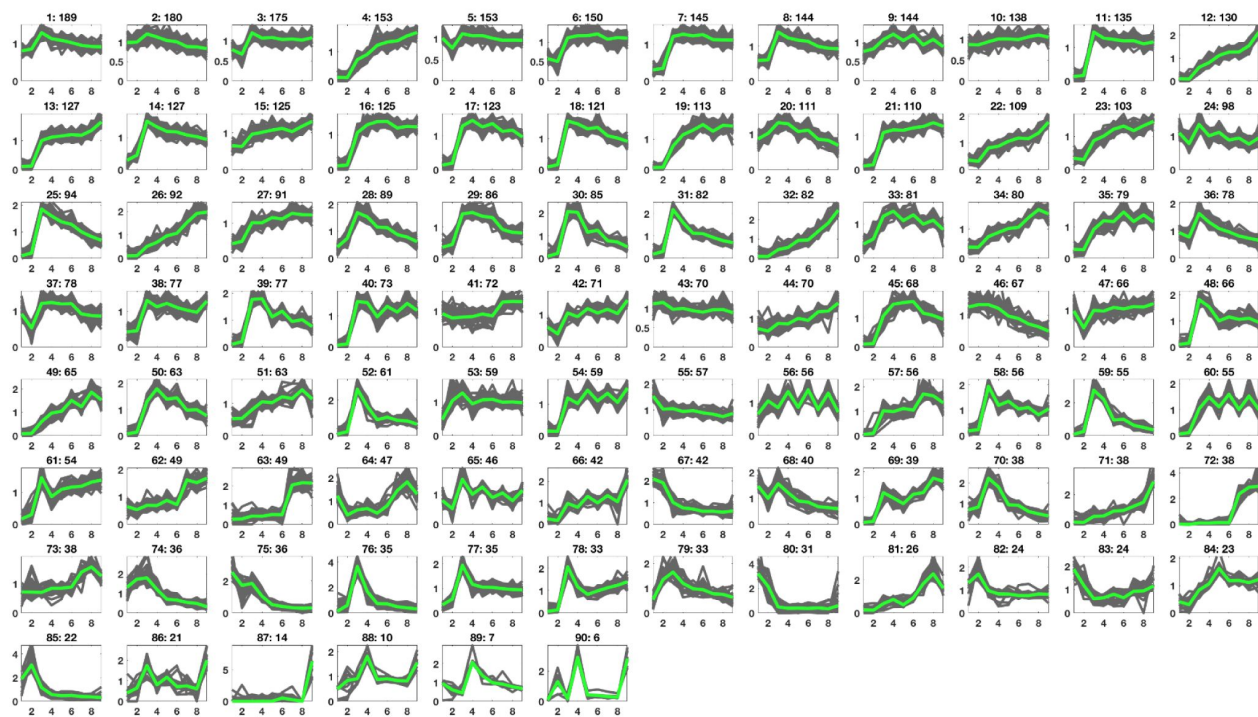
Distribution of absolute protein abundance in the oocyte by functional set. **(A)** Each set is approximated by a Gaussian using respective mean and standard deviation. The number of detected proteins and total proteins in each category is marked in the legend. **(B)** Parameters of Gaussian distributions (mean and standard deviation) for different groups.

Appendix 3

A



B



Appendix 3.

Clusters of dynamics of oocyte protein expression and phosphorylation in response to progesterone. Key clusters of the protein expression (**A**) and phospho-site dynamics (**B**) after progesterone stimulation of an oocyte as obtained by a K-means clustering based on cosine similarity distance. The clusters are arranged in the order of cardinality, starting with the most populated ones. Each subplot is labeled by the cluster ID and the number of proteins in that cluster.

Methods

The relative protein abundance data across subsequent hourly timepoints post- progesterone stimulation was clustered using K-means clustering algorithm with cosine similarity distance. The number of clusters was selected to ensure proteins in each cluster are closely concordant and under-clustered for visualization purpose, resulting in 60 and 90 clusters for protein and phosphopeptides respectively. The code to perform this clustering procedure is available upon request and was run using MATLAB (version R2022b). The K-means clustering algorithm converges to local minima only, therefore, we performed multiple clustering realization to identify the most biologically plausible clusters, as shown in panels A-B. Resulting clusters are ordered by cardinality, most populated clusters first.

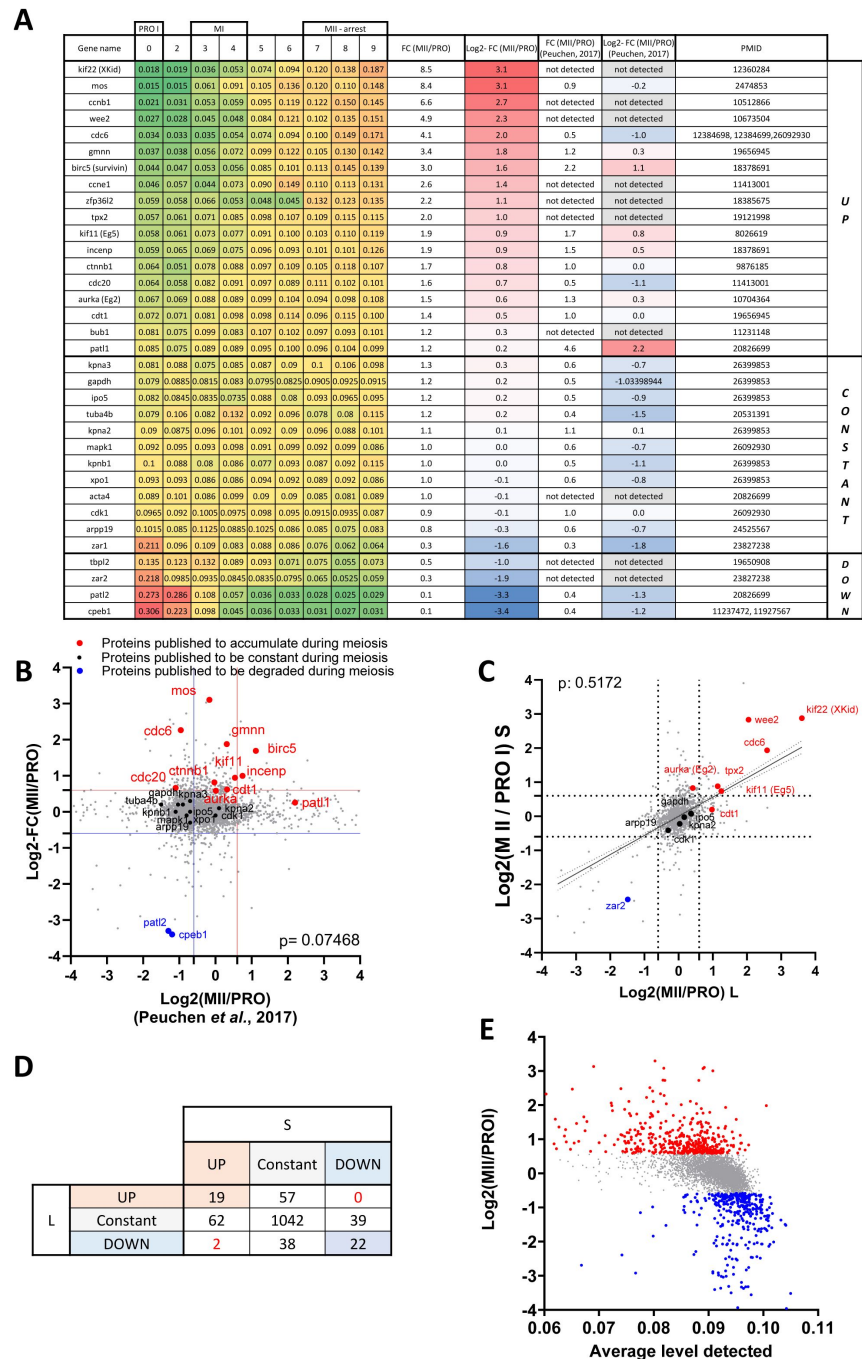


Figure 2-figure supplement 3.

Validation of the proteomic dataset. **(A)** Table summarizing proteins whose changes in protein level have been documented using Western blot. The PMID of the original publication displaying the Western blot is reported. The changes in abundance detected in our dataset are color-coded with a scale from green to red. Log2 fold changes between prophase and metaphase II from our dataset are color-coded with a scale from red to blue and compared to another previously published dataset (Peuchen *et al.*, 2017). **(B)** Proteome-wide comparison of our dataset with (Peuchen *et al.*, 2017). Proteins whose abundance was previously published are color-coded in red for the ones that accumulate, black for the constant, and blue for the ones that are degraded. **(C)** Changes in protein abundance between prophase and metaphase II were calculated for each protein where we could distinguish the .S and .L isoforms, and plotted. The Pearson correlation between the isoforms was calculated with Prism. **(D)** Table summarizing the comparison between .S and .L isoforms presented in panel C. **(E)** Correlation plot between the changes in protein abundance occurring during the prophase to metaphase II transition and the relative abundance in prophase oocytes.

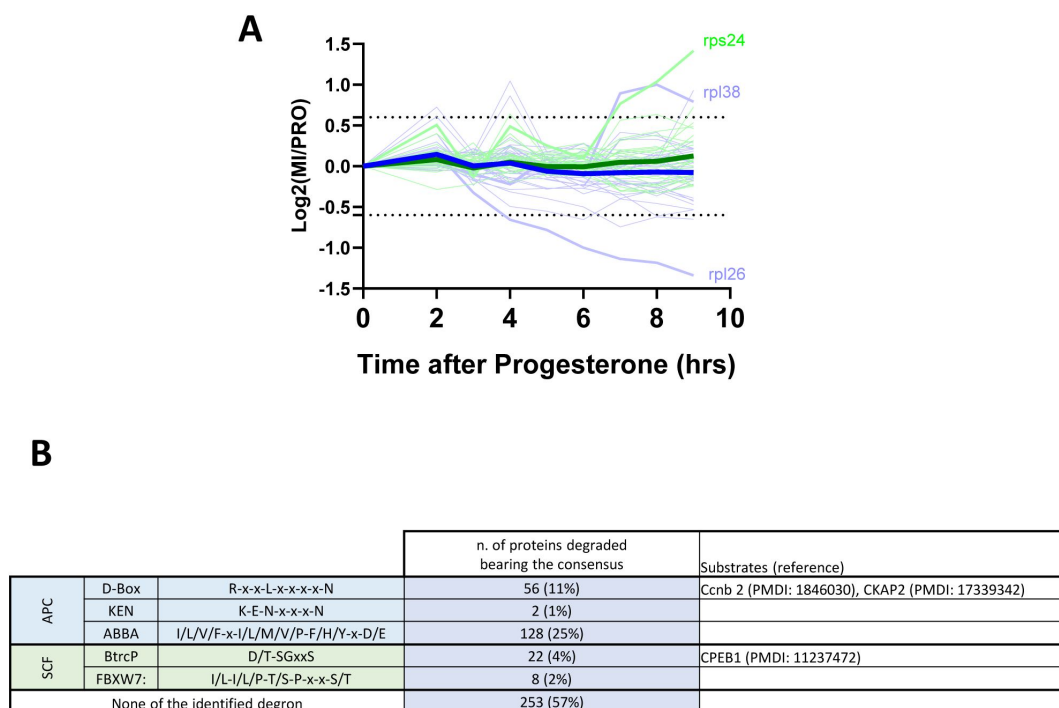


Figure 2-figure supplement 4.

Protein degradation during meiotic maturation: ribosomal proteins, E3 ligases targets. **(A)** Changes in protein abundance of ribosomal proteins between prophase (PRO) and metaphase II (MII) are plotted. Dashed lines represent the threshold used for defining the differentially expressed proteins. **(B)** Table summarizing the different degron consensus motifs identified and the number of proteins in our dataset bearing these motifs. Examples of proteins bearing different degrons are listed along with supporting literature.

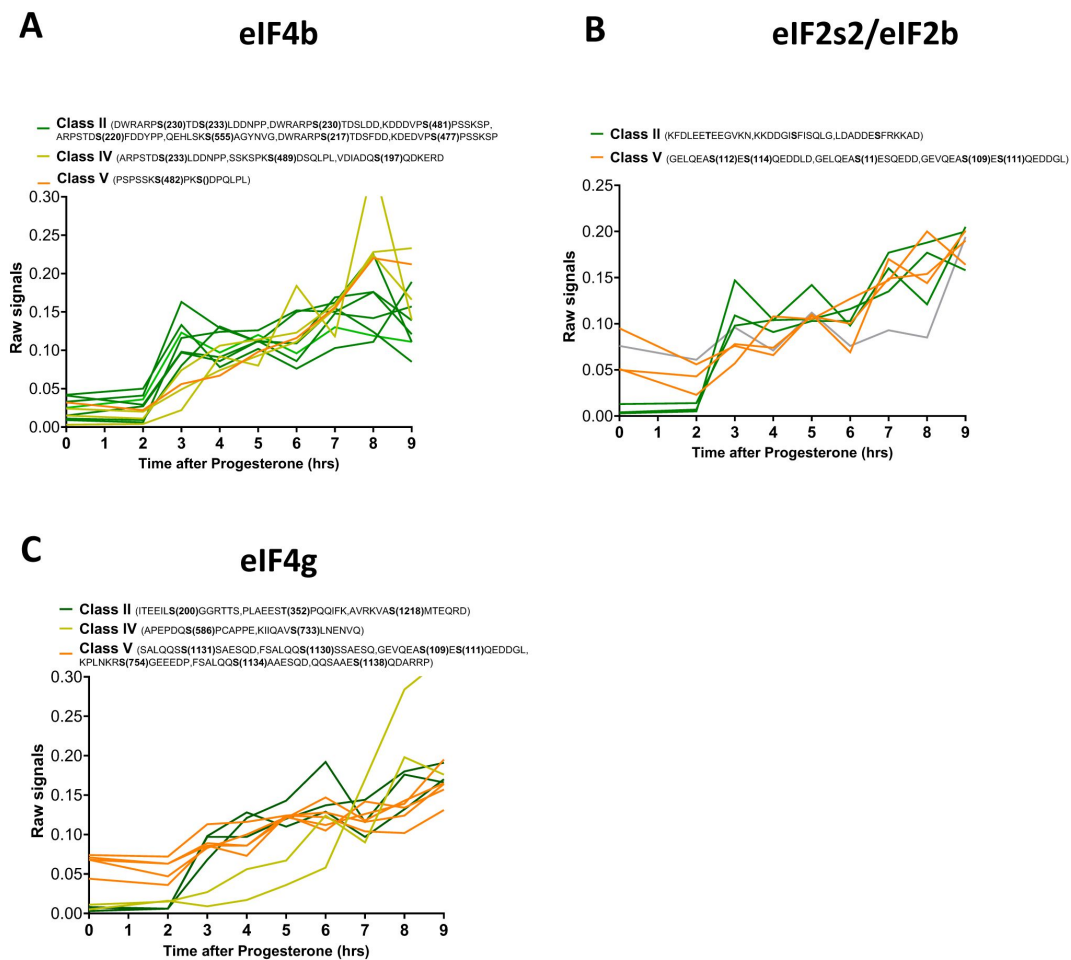


Figure 4-figure supplement 1.

Phosphorylation patterns of components of the initiation translation machinery during meiotic divisions. (**A-C**) Levels of phosphorylation sites of eIF4b (**A**), eIF2s2/eIF2b(**B**) and eIF4g (**C**) during meiotic divisions are plotted. The phosphorylation curves are color-coded: Class II, green; Class IV, yellow, Class V, orange.

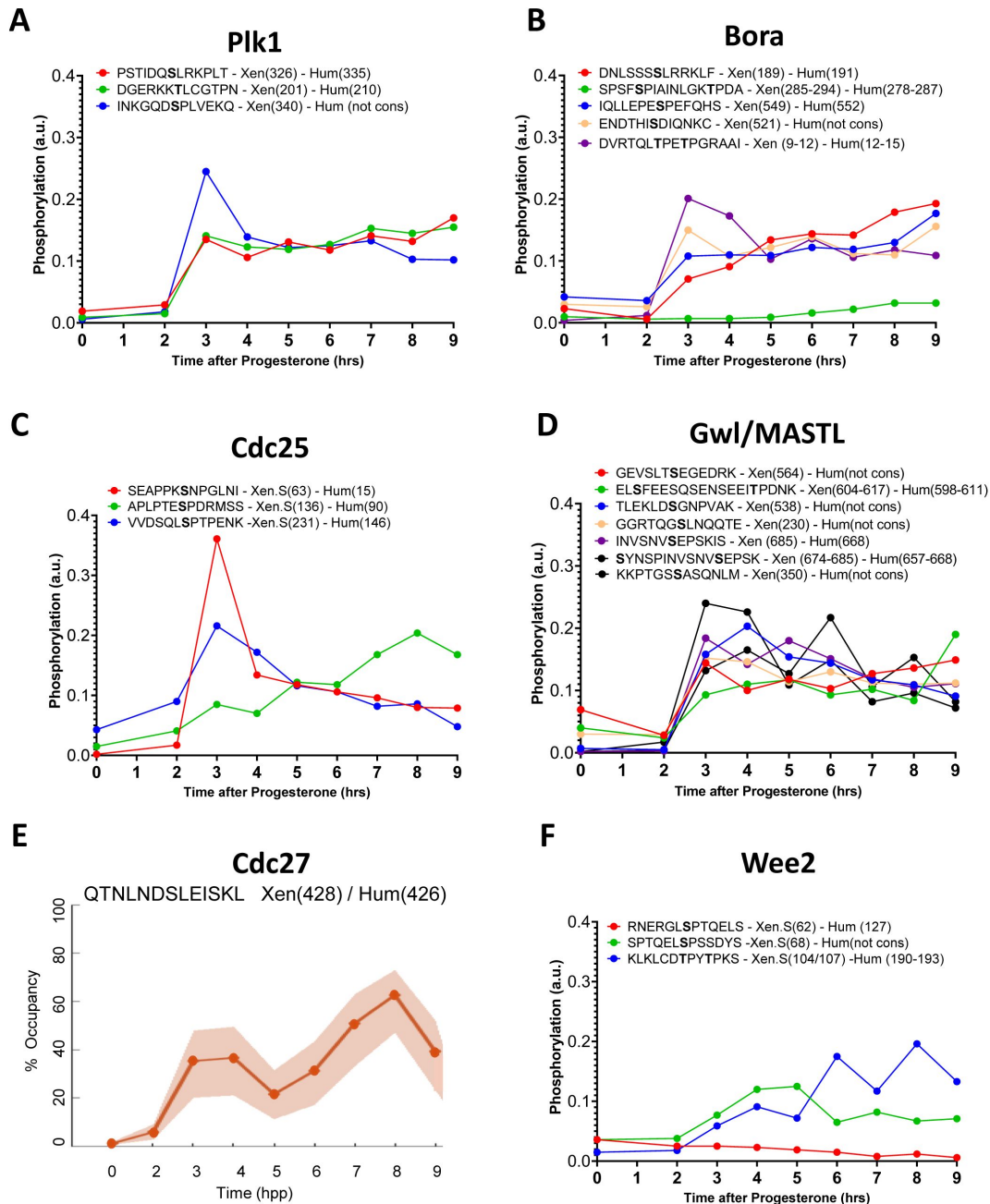


Figure 6-figure supplement 1.

Phosphorylation patterns of Cdk1 regulators during meiotic divisions. (**A-D** and **F**) Levels of phosphorylation sites of Plk1 (**A**), Bora (**B**), Cdc25 (**C**), Gwl/MASTL (**D**) and Wee2 (**F**) during meiotic divisions are plotted. The peptide sequences are reported with the phosphorylation sites marked in bold. The number of the residue phosphorylated is reported between breakers according to the *Xenopus* and human nomenclature. (**E**) The phosphorylation occupancy for S428 of Cdc27 is plotted with its confidence intervals.

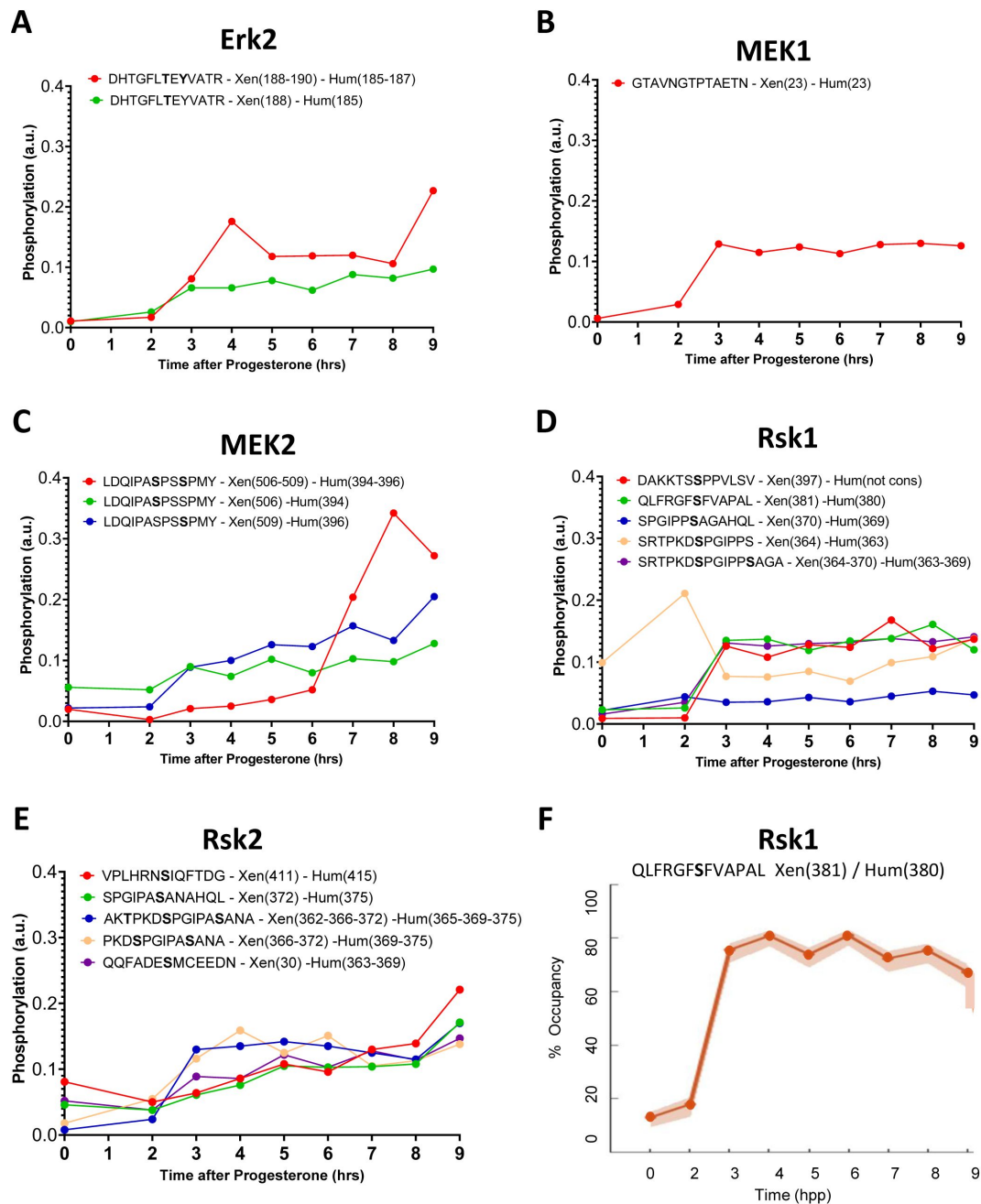


Figure 6-figure supplement 2.

Phosphorylation patterns of the mos/MAPK pathway during meiotic divisions. **(A-E)** Levels of phosphorylation sites of Erk1 **(A)**, Bora **(B)**, MEK1 **(C)**, MEK2 **(D)** and Rsk2 **(E)** during meiotic divisions are plotted. The peptide sequences are reported with the phosphorylation sites marked in bold. The number of the residue phosphorylated is reported between breakers according to the *Xenopus* and human nomenclature. **(F)** The phosphorylation occupancy for S381 of Rsk1 is plotted with its confidence intervals.

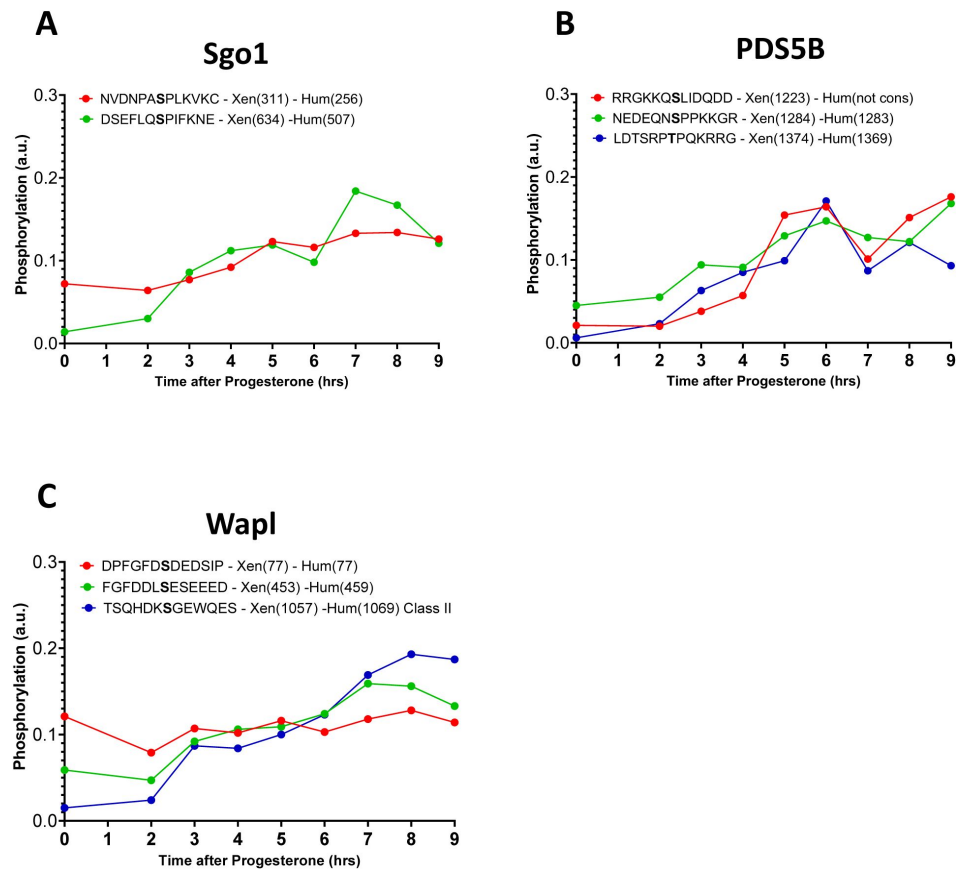


Figure 8-figure supplement 1.

Phosphorylation patterns of cohesins during meiotic divisions. **(A-C)** The levels of phosphorylation sites of Sgo1 **(A)**, PDS5B **(B)** and Wapl **(C)** during meiotic divisions are plotted. The peptide sequences are reported with the phosphorylation sites marked in bold. The position of the residue phosphorylated is reported in brackets according to the *Xenopus* and human nomenclature.

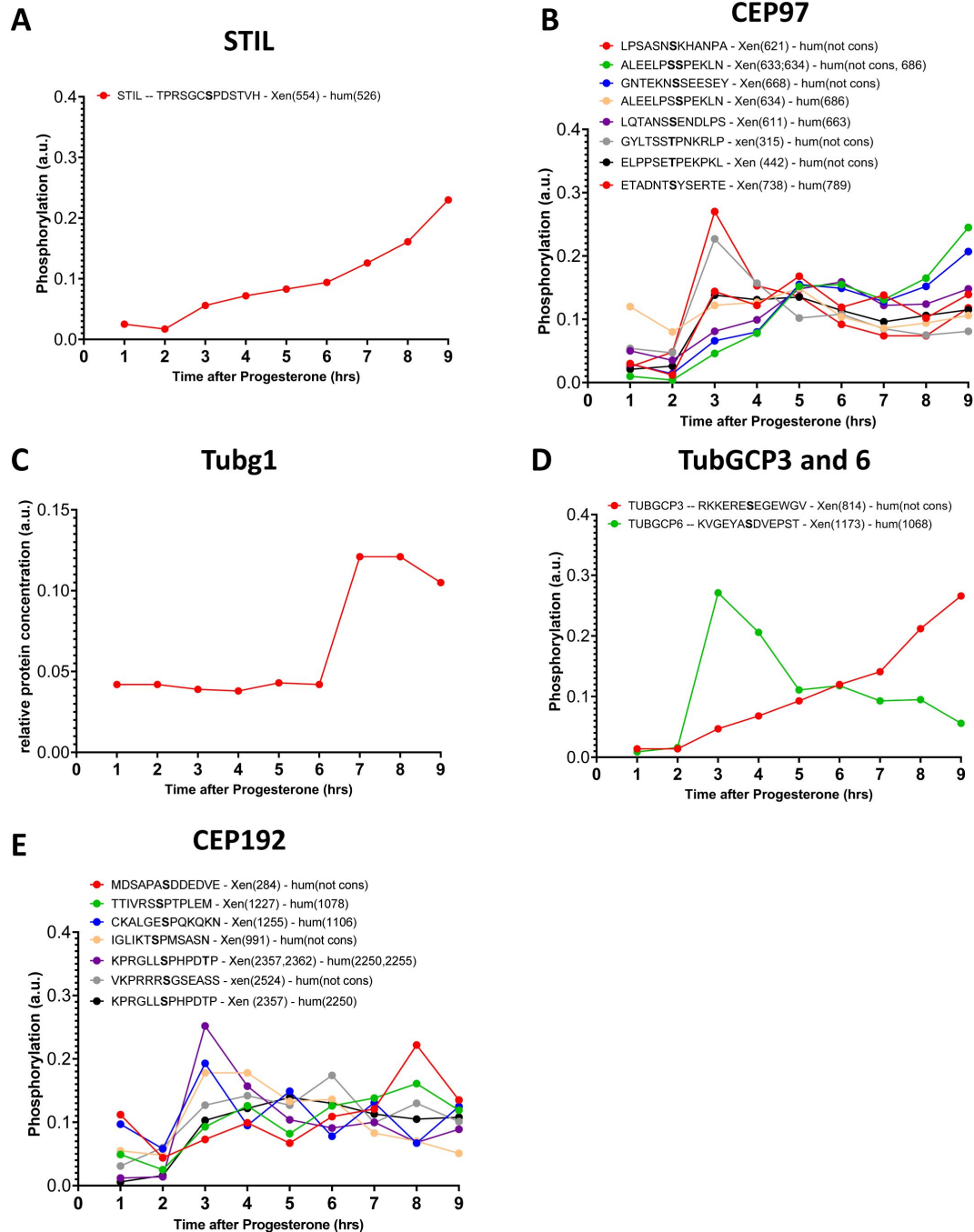


Table 1-table supplement 1.

Phosphorylation patterns of centrosomal proteins during meiotic divisions.

(A-E) The levels of phosphorylation sites of STIL (A), CEP97 (B), Tubg1 (C), TubGCP3 and 6 (D), CEP192 (E) during meiotic divisions are plotted. The peptide sequences are reported with the phosphorylation sites marked in bold. The number of the residue phosphorylated is reported between breakers according to the *Xenopus* and human nomenclature.

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Reviewer #1 (Public review):

Summary:

The study aims to create a comprehensive repository about the changes in protein abundance and their modification during oocyte maturation in *Xenopus laevis*.

Strengths:

The results contribute meaningfully to the field.

Weaknesses:

The manuscript could have benefitted from more comprehensive analyses and clearer writing. Nonetheless, the key findings are robust and offer a valuable resource for the scientific community.

<https://doi.org/10.7554/eLife.104255.1.sa3>

Reviewer #2 (Public review):

Summary:

The authors analyzed *Xenopus* oocytes at different stages of meiosis using quantitative phosphoproteomics. Their advanced methods and analyses revealed changes in protein abundances and phosphorylation states to an unprecedented depth and quantitative detail. In the manuscript they provide an excellent interpretation of these findings putting them in the context of past literature in *Xenopus* as well as in other model systems.

Strengths:

High quality data, careful and detailed analysis, outstanding interpretation in the context of the large body of the literature.

Weaknesses:

Merely a resource, none of the findings are tested in functional experiments.

I am very impressed by the quality of the data and the careful and detailed interpretation of the findings. In this form the manuscript will be an excellent resource to the cell division community in general, and it presents a very large number of hypotheses that can be tested in future experiments.

Xenopus has been and still is a popular and powerful model system that led to critical discoveries around countless cellular processes, including the spindle, nuclear envelope, translational regulation, just to name a few. This also includes a huge body of literature on the cell cycle describing its phosphoregulation. It is indeed somewhat frustrating to see that these earlier studies using phospho-mutants and phospho-antibodies were just scratching the surface. The phosphoproteomics analysis presented here reveals much more extensive and much more dynamic changes in phosphorylation states. Thereby, in my opinion, this manuscript opens a completely new chapter in this line of research, setting the stage for more systematic future studies.

<https://doi.org/10.7554/eLife.104255.1.sa2>

Reviewer #3 (Public review):**Summary:**

The authors performed time-resolved proteomics and phospho-proteomics in *Xenopus* oocytes from prophase I through the MII arrest of the unfertilized egg. The data contains protein abundance and phosphorylation sites of a large number set of proteins at different stages of oocyte maturation. The large sets of the data are of high quality. In addition, the authors discussed several key pathways critical for the maturation. The data is very useful for the researchers not only researchers in *Xenopus* oocytes but also those in oocyte biology in other organisms.

Strengths:

The data of proteomics and phospho-proteomics in *Xenopus* oocyte maturation is very useful for future studies to understand molecular networks in oocyte maturation.

Weaknesses:

Although the authors offered molecular pathways of the phosphorylation in the translation, protein degradation, cell cycle regulation, and chromosome segregation. The author did not check the validity of the molecular pathways based on their proteomic data by the experimentation.

<https://doi.org/10.7554/eLife.104255.1.sa1>

Author response:

We are both honored and humbled by the high praise our work received from all three reviewers. Below, we address the common comments made by the reviewers:

(1) Value and Impact of the Resource: We are grateful for the recognition of our dataset as a valuable and high-quality resource. Our primary goal was to generate a comprehensive dataset on protein abundance and phosphorylation dynamics during *Xenopus* oocyte maturation. We are pleased that this work has been seen as a solid foundation for future studies in *Xenopus* research and beyond, with broader implications for oocyte and cell cycle biology.

(2) Focus on Functional Validation and Contextualization with Prior Studies: The manuscript was submitted as a Tools and Resources article, a format that emphasizes the creation and presentation of datasets, tools, and methodological advances to facilitate future discoveries. In alignment with this format, we ensured that the information is accessible and deployable for the broader scientific community. While we did not include functional validation of specific pathways, the dataset provides a robust framework for generating numerous testable hypotheses. We plan to pursue some of these follow-up studies in our labs and encourage the community to explore these further.

(3) Contextualization with Prior Studies: We appreciate the recognition of our efforts to integrate our findings with the existing body of literature. In conclusion, we would like to thank the reviewers for their evaluation and thoughtful suggestions. We look forward to seeing how this dataset contributes to future discoveries in the field.

<https://doi.org/10.7554/eLife.104255.1.sa0>